

Predator SS

SIMPLIFIED STRETCH

****CSA Approved****

LOW PROFILE

SERIAL NUMBER

Please refer to the serial number in all correspondence with Highlight or any Highlight Distributor. This identifies your machine and will help in our ability to quickly and efficiently respond to your needs.

HIGHLIGHT
INDUSTRIES, INC.

OPERATION MANUAL NUMBER: PREDATOR SS CSA-CUL/2023

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Table of Contents

Table of Contents	1
Table of Figures.....	2
General Information	3
Safety Messages	4
Introduction and Warranty	6
Revision History.....	8
Specifications.....	9
Specifications	10
System Overview Prints.....	11
System Description.....	12
Installation & Adjustments.....	13
Machine Placement	14
Machine Set-Up.....	14
Optional Scale Package	18
Operation Instructions.....	21
Operator Controls	22
Film Loading	24
Machine Operating Instructions	26
Maintenance	27
Preventative Maintenance	28
Maintenance Schedule	31
Trouble-shooting.....	32
Turntable Problems	33
Film Carriage Problems	34
Technical References	35
Recommended Spare Parts	36
Electrical Drawings	37
Mechanical Drawings.....	51
Notes	66

Table of Figures

Figure 1: Predator SS Low Profile Overview Layout	11
Figure 2: Predator SS Low Profile Machine Description.....	12
Figure 3: Bracket in the Shipping Position.....	15
Figure 4: Bracket in its Operating Position	16
Figure 7: Film Loading/Threading Diagram	24
Figure 8: Turntable Belt Adjustment.....	29
Figure 9: Film Carriage Lift Chain Adjustment.....	30

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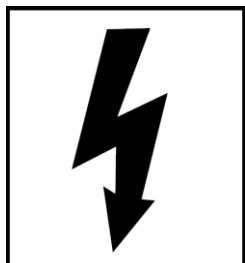
Predator SS Turntable Stretch Wrapper Operation Manual

General Information

Safety Messages

For the best result with the Predator Turntable Stretch Wrapper, carefully read this manual and all of the warning labels attached to the equipment before installing and operating it, and follow instructions exactly. Keep this manual for machine reference.

Definitions and Symbols



High Voltage!

This symbol indicates high voltage. It calls your attention to items or operations that could be dangerous to you or other persons operating the equipment. Read the message and follow the instructions carefully.



Warning!

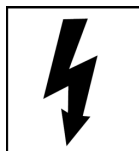
This symbol indicates a potentially hazardous situation which, if not avoided, can result to bodily injury, or serious damage to the product.



Notes

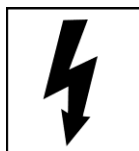
This symbol indicates an area or subject of special merit, emphasizing the equipment's capabilities, common errors in operation or maintenance, or other special instructions that can provide benefits to users.

General Precautions – Read These First!



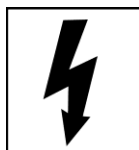
High Voltage!

Disconnect AC input power before checking components, performing maintenance, cleaning up, and when the machine is not in use. Do NOT connect or disconnect wires and connectors while power is applied to circuit.



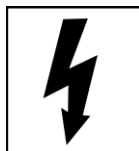
High Voltage!

Wiring work should be carried out only by qualified personnel. Otherwise, there is a danger of electric shock or fire.



High Voltage!

The user is responsible for conforming to all applicable code requirements with respect to grounding all requirements. Do NOT use extension cords to operate the equipment.



High Voltage!

Motor control equipment and electronic controllers are connected to hazardous line voltages. When servicing drives and controllers, there may be exposed components with housings or protrusions at or above line potential. Extreme care should be taken to protect against shock.



Warning!

Loose clothing must NOT be worn while the machine is in operation. Stay clear of moving parts while the machine is running.

Introduction and Warranty

Introduction

Thank you for your purchase of a *Predator SS* Turntable Stretch Wrapper! Designed for the high volume industrial user the Predator SS will stretch wrap any load to the maximum stretch level of any film with full independent control of the film force. This allows even light unstable loads to be wrapped at the lowest possible cost and the maximum load holding force.

Limited Warranty

Highlight Industries, Inc. warrants its Predator manufactured by it, and sold pursuant to this order, will be of merchantable quality, free from defects in material and workmanship as determined at the date of shipment, by generally recognized, applicable and accepted practices and procedures in the industry, for a period of three (3) years from the Highlight invoice date, under normal use and service.

When the Purchaser gives Highlight written notice of any alleged defect within the applicable warranty period, Highlight will, at its option repair or replace the same free of charge F.O.B. its manufacturing plant, installation not included. Equipment replaced under the warranty shall have the same warranty as new equipment but does not extend the warranty of the original equipment.

Satisfaction of this warranty, consistent with other provisions herein, will be limited to the replacement or repair or modification of, or issuance of a credit for, the equipment involved, at Highlight's option.

Highlight neither assumes nor authorizes any person to assume for it any other obligation in connection with the sale of Highlight's equipment.

This warranty shall not apply to any equipment which has been repaired or altered by unauthorized personnel in any way so as to, in the judgment of Highlight, affect serviceability, or which has been subjected to misuse, negligence, accident, or to equipment made by Highlight which has been operated in a manner contrary to Highlight's instructions.

In no event regardless of the cause, shall Highlight be liable for penalties or penalty clauses of any description or any damages resulting from loss of profits, use of products or for any incidental indirect or consequential damages, even if advised of the possibility of such damages. This limitation of Highlight's liability will apply regardless of the form of action, whether in contract or tort, including negligence. Any action against Highlight must be brought within twelve (12) months after cause of action accrues.

"This warranty in lieu of all other warranties whether expressed, implied or statutory including implied warranties of merchantability of fitness or extends only to the buyer or customer purchasing from Highlight Industries, Inc. or an authorized distributor."

About This Document

The purpose of this manual is to provide you with information necessary to install, operate, troubleshoot, and maintain the Predator SS Turntable Stretch Wrapper. The audience for this manual should have basic knowledge of mechanical and electronic components, standard electrical wiring practices, and schematics symbols.

To guarantee safe operation of the equipment, carefully observe the safety messages throughout this manual. Keep this operating manual and distribute it to all users for reference.

Application Assistance

If any assistance is desired, contact the distributor from whom you have purchased the unit, or call the number listed on the bottom of each page of this manual. To receive quick and proper technical support for the equipment you have purchased, please be prepared to provide the following information:

1. Machine Serial Number
2. Date of Purchase
3. Symptoms of any problems

Revision History

Revision	Revision Comment	Date of Revision
	Initial Release	June 2022

Highlight Industries, Inc.

Predator SS Turntable Stretch Wrapper Operation Manual

Specifications

Specifications

Machine Dimensions

	Low Profile
Length (inch)	98
Width (inch)	60
Height (inch)	91
Turntable Diameter (inch)	60
Turntable Height from Floor (inch)	3.25
Operation Space (inch)	105 x 75 x 120
Maximum Pallet Size (inch)	52 x 52 x 72
Shipping Weight (lbs)	1050

Electrical Specifications

- 120VAC, 60Hz, Single Phase, 15Amp

Turntable

- 4,000 lbs Maximum Turntable Load Capacity
- 12 RPM Maximum Turntable Speed
- 20-30 Loads Per Hour

Film Carriage

- Adjustable Raise and Lower Speeds
- Automatic Package Height Detection

Film Delivery

- Infinite/Manual Film Force Adjustment
- 10 Inch Diameter Roll Capacity
- 20 Inch Height Roll Capacity (30 Inch Optional)

System Overview Prints

**Note**

Shown below are STANDARD assembly drawings. It may not reflect your purchased system, especially when optional items are added. Refer to the Technical References section of the manual for more detailed information.

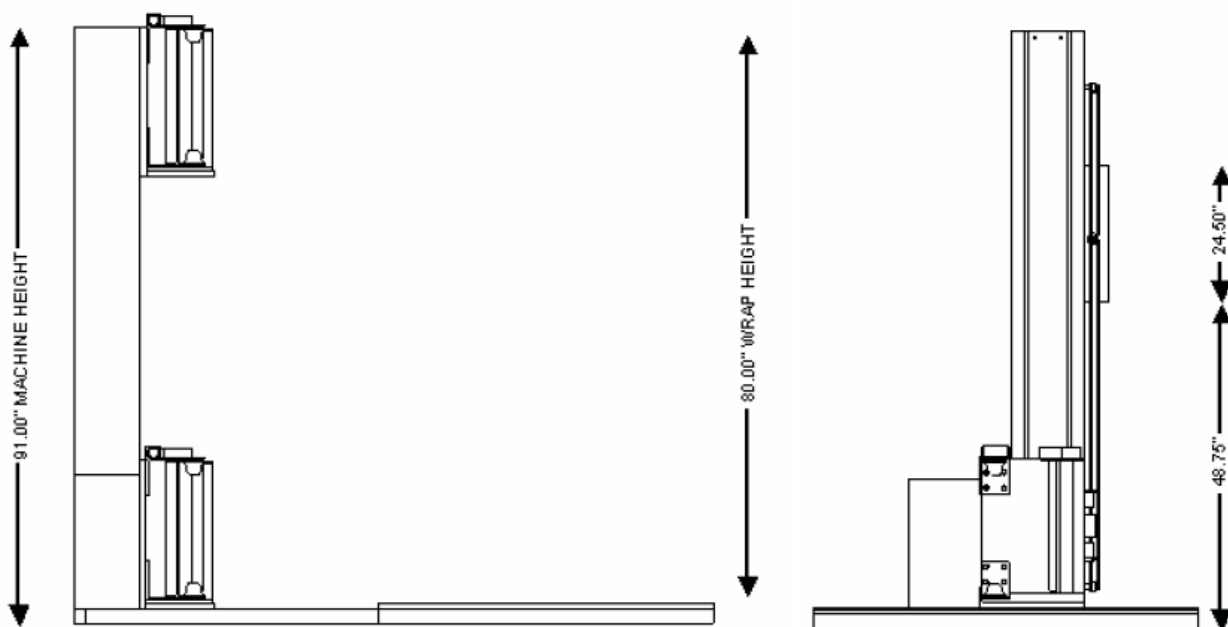
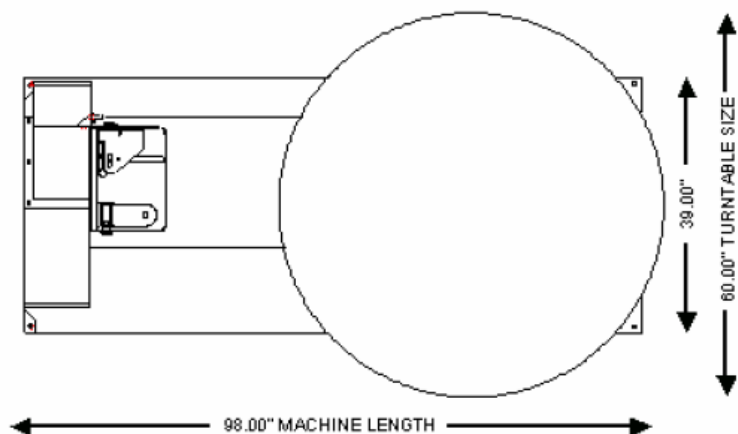


Figure 1: Predator SS Low Profile Overview Layout

System Description

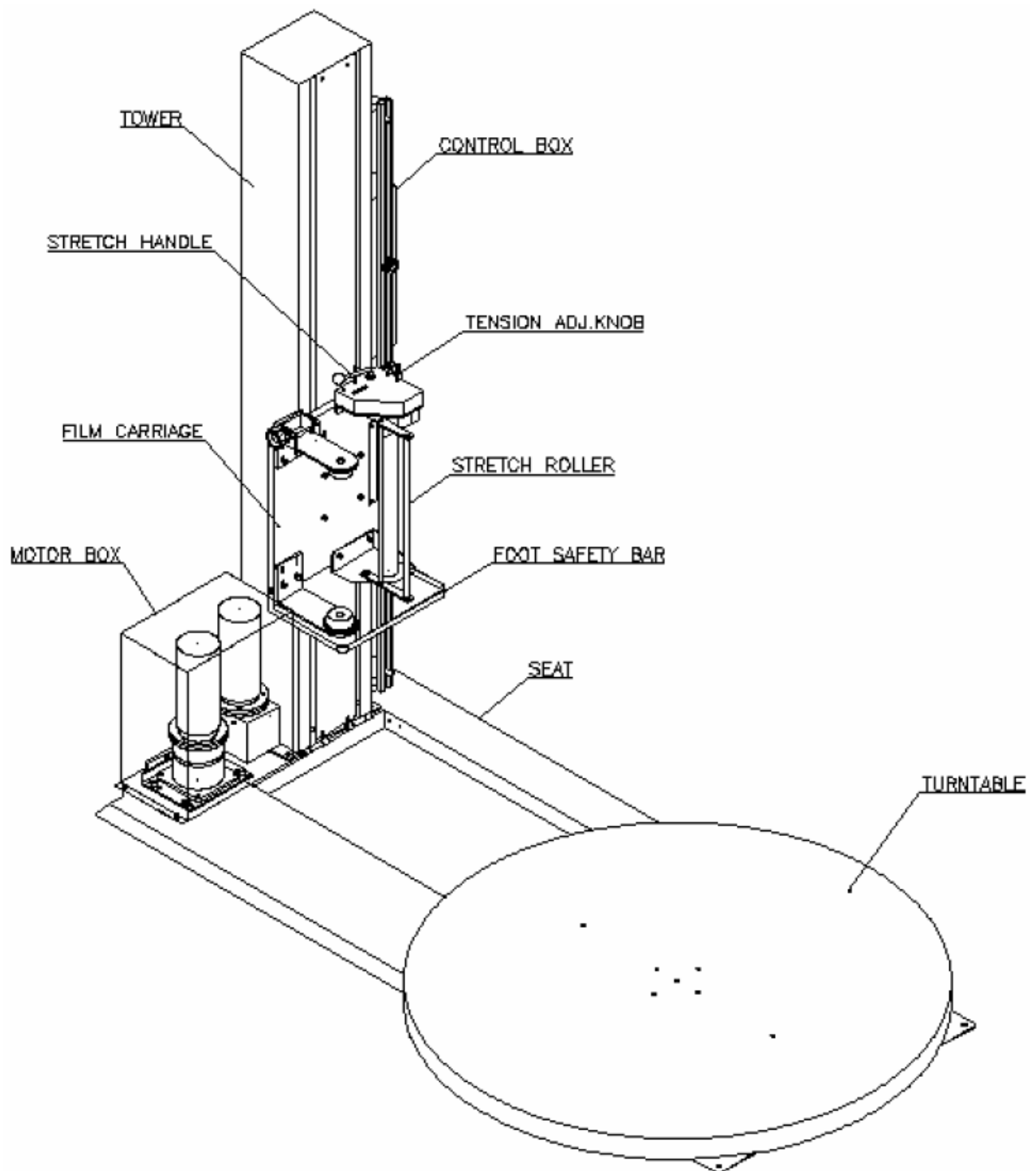


Figure 2: Predator SS Low Profile Machine Description

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Predator SS Turntable Stretch Wrapper Operation Manual

Installation & Adjustments

Machine Placement

Receiving and Inspections

The Predator Turntable Stretch Wrapper has gone through quality control tests at the factory before shipment. Upon receiving, please do the following:

1. Inspect the entire machine for visual damage during shipment. If found, immediately report this damage to the trucking line. *Highlight Industries, Inc.* has taken every precaution during the packaging and loading of this equipment, however, it is YOUR RESPONSIBILITY to inspect for damage before installation.
2. Make sure the part or serial number indicated on the machine corresponds with the part number of your order.

Positioning of the Machine

Place the Predator Turntable Stretch Wrapper close to an area where you will be wrapping your loads. Make sure that there is sufficient room to load/unload the machine and that you do not stretch the power cable. Remember, you will need to plug the machine in to a 120VAC, 15-Amp outlet.

Floor Weight Bearing/Stress Tolerance

The floor must be able to bear the weight of the machine, the weight of the maximum load, plus a safety factor. The floor must also be able to tolerate the stress of the machine's operation. If the fork trucks will operate on the same weight bearing area, add the weight of the trucks to the weight bearing stress tolerance requirements.

The Predator can be installed on any type of floor constructions that meets the weight bearing and stress tolerance requirements.



Warning!

Highlight does not recommend anchoring machines to the floor unless there is a critical space concern or other concern about the machine moving when bumped (such as near an aisle way or other machine). Allowing the machine to move slightly when bumped greatly reduces the likeliness of damage as it prevents the impact from going directly into the turntable/frame.



Warning!

Do **NOT** anchor machines with scale options to the floor as this will result in calibration issues, incorrect weights, and possible damage to scale components. If securing the machine is required, please contact your distributor to obtain securing floor plates.

Machine Set-Up

Unpacking and Moving the Machine

It is very important to read all instructions before undertaking any of these steps. The following steps should help achieving a safe and quick machine set-up.

1. Place the skidded machine close to the designated wrap area. Remove all shipping fasteners holding the machine to the pallet. The machine may be crated with the tower tilted down with the motor cover front carriage roller removed for shipping purposes.
2. Place the forks of the forklift through the tubes provided on the base of the machine, remove the machine from the shipping skids, and place it in the designated wrap area.
3. If the OPTIONAL ramp (Part Number #600086) is purchased:
Select a ramp position as illustrated below. The ramp can be positioned anywhere in a 180° rotation around the front of the turntable. There should be ¼" gap between the turntable and the ramp. The ramp should be fully supported by the floor. Both the ramp and the machine should be lagged to the floor.
4. When a machine is ordered with a 110" extended tower or a scale package, the carriage photo eye bracket will need to be unbolted from its shipping position (Figure 3) to its operating position (Figure 4).



Figure 3: Bracket in the Shipping Position



Figure 4: Bracket in its Operating Position



Warning!

It is very important that the machine be leveled. Uneven floors will cause premature turntable support roller failure.

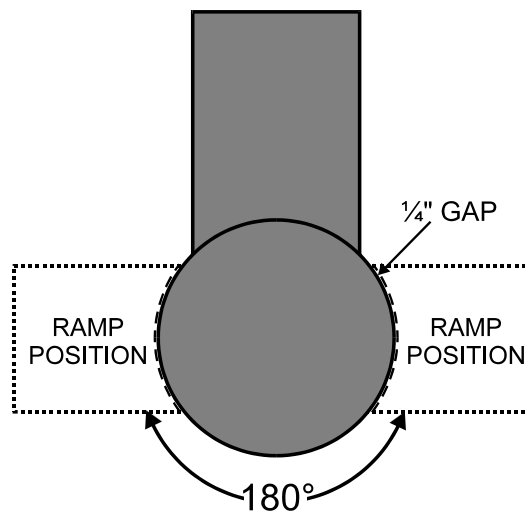


Figure 5: Ramp Position for the Predator SS Low Profile

Power and Control Wiring Checks

1. Using a volt meter, check the AC voltage coming to the machine and insure the proper voltage is present.
2. Make sure the Emergency Stop button on the operator panel is fully pressed in. Turn the disconnect switch to the ON position. Power should be applied to the drive board, operating touch-panel, photoelectric sensors, switches, and LED's.
3. Press the Emergency Stop button. Make sure all machine power is completely removed when the Emergency Stop button is depressed. Pull the Emergency Stop button out to resume.
4. Open the electric control box. Make sure all machine power is completely removed when the electrical control box is open. Close the electrical control box to resume.
5. Trip the carriage foot security bar. Make sure all machine power is completely removed when the carriage foot security bar is tripped. Clear the bar to resume.

**Warning!**

Ensure that the Emergency Stop button light is ON when the button is pushed in and OFF when the button is pulled out.

Optional Scale Package

The optional scale package allows the user to quickly measure the weight of the package placed on the machine. To operate correctly and efficiently the scale needs to be configured with the correct settings. Below are the instructions on how to re-configure the scale and the settings that have been pre-programmed. Re-configuring the scale is not required unless there is a problem with the existing scale or a new scale is added to the machine. Refer to the scale manual for more information.

**Note**

Changing the scale parameters to values other than the Highlight recommended values may result in poor scale operation.

Navigating the Scale Menu

Step 1: Turn the scale display over so you are looking from above down at the back side of the scale and turn off the scale power.

Step 2: Remove the two small screws holding a small plate on the back of the scale.

Step 3: Looking down at the scale, move the switch under the plate to the right.

Step 4: Turn the scale power back on.

Step 5: Press the “PRINT / >>” button until the desired setting appears on the screen.

Step 6: Press the “ZERO” button to enter the parameter set-up.

Step 7: Press the “PRINT / >>” button until the display reads the correct setting and press the “SET / NET” button.

Step 8: Turn the power back off, move the switch under the plate to the left, and turn the power back on.

Step 9: Replace the screws on the back cover and retighten the scale swivel grips on the side.

Step 10: The scale is now ready to use.

Highlight Recommended Configuration Values:



Note

Changing the scale parameters to values other than the Highlight recommended values may result in poor scale operation.

Name/Code	Description	Highlight Recommended Value
F1: Graduations	Specifies the number of full-scale graduations. Value should be consistent with legal requirements and environmental limits on the useful system resolution.	5000
F2: Span Gain	Span Gain is related to A/D integration time. The larger the span gain, the higher the internal resolution, but the slower the update speed. Note that the scale must be re-calibrated whenever this parameter is altered. See Appendix C in the scales manual for more information.	100
F3: Zero Track Band	Selects the range within which the scale will automatically zero. Note that the scale must be in a standstill to zero the scale.	3
F4: Zero Range	Selects the range within which the scale may be zeroed. Note that the indicator must be in standstill to zero the scale.	100
F5: Motion Band	Sets the level at which motion is detected by comparing the present display update with the previous one. If motion is not detected for two seconds or more, scale is in standstill and can process a Print or Zero command. Maximum value varies depending on local regulations	1
F6: Digital Filter	Averages weight readings to produce higher stability but the slower the indicator's response time. Choose 8 unless a very fast response is needed.	8
F7: Overload Limit	Selects the desired formula which determines the point at which the indicator shows overload. All selections are based on the primary unit selected in F8.	2%
F8: Calibration Unit	Selects the primary base unit to be used in the calibration process. Also the default unit for normal operation. 1 = pounds, 2 = kilograms.	1
F9: Display Divisions	Determines the desired weight increments. Value should be consistent with legal requirements.	1
F10: Decimal Point	Determines the location of the decimal point.	0
F16: Zero Calibration	Places the scale in a Zero Calibration routine.	See below.
F17: Span Calibration	Places the scale in a Span Calibration routine. See below.	See below.

Scale Calibration Procedure



Note

Highlight Industries, Inc calibrates the scale before shipment using known weights. The values stored in these settings are custom per scale and machine. Do not change these values unless directed to do so by Highlight personnel.

Step 1: Navigate the scale menu to parameter “F 16.”

Step 2: Press the “ZERO” button ONCE. This enters the Zero Calibration Menu. The display will momentarily show “C 0” followed by a value.

Step 3: After making sure that there are no test weights on the machine press the “ZERO” button again. The display will zero.

Step 4: Press the “SET / NET” button to save the zero point value. The display will show “EndC0” momentarily then revert to parameter F16.

Step 5: Navigate to parameter “F 17.”

Step 6: Press the “ZERO” button ONCE. This enters the Span Calibration Menu. The display will momentarily show “C 1” followed by a value with one flashing digit. This value will be zero with the decimal point parameter selected in F10.

Step 7: Place a known weight on the machine. Ensure that the test weight is below the maximum capacity of the scale and above 1% of the maximum capacity of the scale.

Step 8: Key in the weight of the test weight using the buttons on the screen. Press the “TARE / <<” button to select the digit to the left of the current position. Press “PRINT / >>” to select the digit to the right of the current position. Press “UNITS” to increase the value of the digit and press “ZERO” to decrease the value of a digit.

Highlight Industries, Inc.

Predator SS Turntable Stretch Wrapper Operation Manual

Operation Instructions

Operator Controls



Warning!

If the Emergency Stop button is pressed while the turntable is rotating, the turntable will NOT stop immediately, but rather it will decelerate and possibly coast to a stop.

The operator panel for the Predator SS is shown and described as follows:

Operator Panel Button Controls



Start

The “Start” button initiates all operations. The button LED stays on until the wrapping cycle is complete. In Machine Reset mode, the button LED flashes until complete.

Mode/Reset

Switch to “Reset” and press the “Start” button to clear any operation. The film carriage will lower to the bottom limit switch and the turntable will return to its home position.

Switch to “Top Wraps” to select Top Wraps First mode. Press the “Start” button to begin cycle. The film carriage will begin traveling upwards to top of the product, apply top wraps, travel downwards to the bottom limit, and apply the bottom wraps to finish the cycle.

Switch to “Bottom Wraps” to select Bottom Wraps First mode. Press the “Start” button to begin cycle. The film carriage will begin applying the bottom wraps, travel upwards to top of the package, apply top wraps, and travel downwards to finish cycle.

Top Wraps

Switch to the desired number of rotations for applying wraps to the top of the product.

Bottom Wraps

Switch to the desired number of rotations for applying wraps to the bottom of the product.

Turntable

The Turntable Jog/Home button manually jogs the turntable while the button is being pressed. To send the turntable to home, press the button twice in quick succession.

Film Carriage

To raise or lower the film carriage manually, switch and hold in the “Manual Up” or “Manual Down” position. Release the switch to stop.

Turntable Speed

The “Turntable Speed” potentiometer dial determines the speed of the turntable. Turn clockwise to increase, counter-clockwise to decrease. Adjusting this will affect the film overlap. The maximum turntable speed is 12 revolutions-per-minute (RPM).

Carriage Speed

The “Carriage Speed” potentiometer dial determines the speed of the carriage. Turn clockwise to increase, counter-clockwise to decrease. Adjusting this will affect the film overlap.

Stop

The “STOP” button cuts machine operation and removes power to frequency drives. In the event this button is pressed during the course of operation, it is necessary to pull this button fully out to reset the machine.

Film Loading

Follow the procedure below to thread the film onto the Film Carriage.

**Warning!**

Be sure the Emergency Stop button is pressed before threading the film and while the film is being threaded.

1. Place the film on the film mandrel.
2. Set the Stretch Handel to the **OFF** position to release the tension. (See figure below)

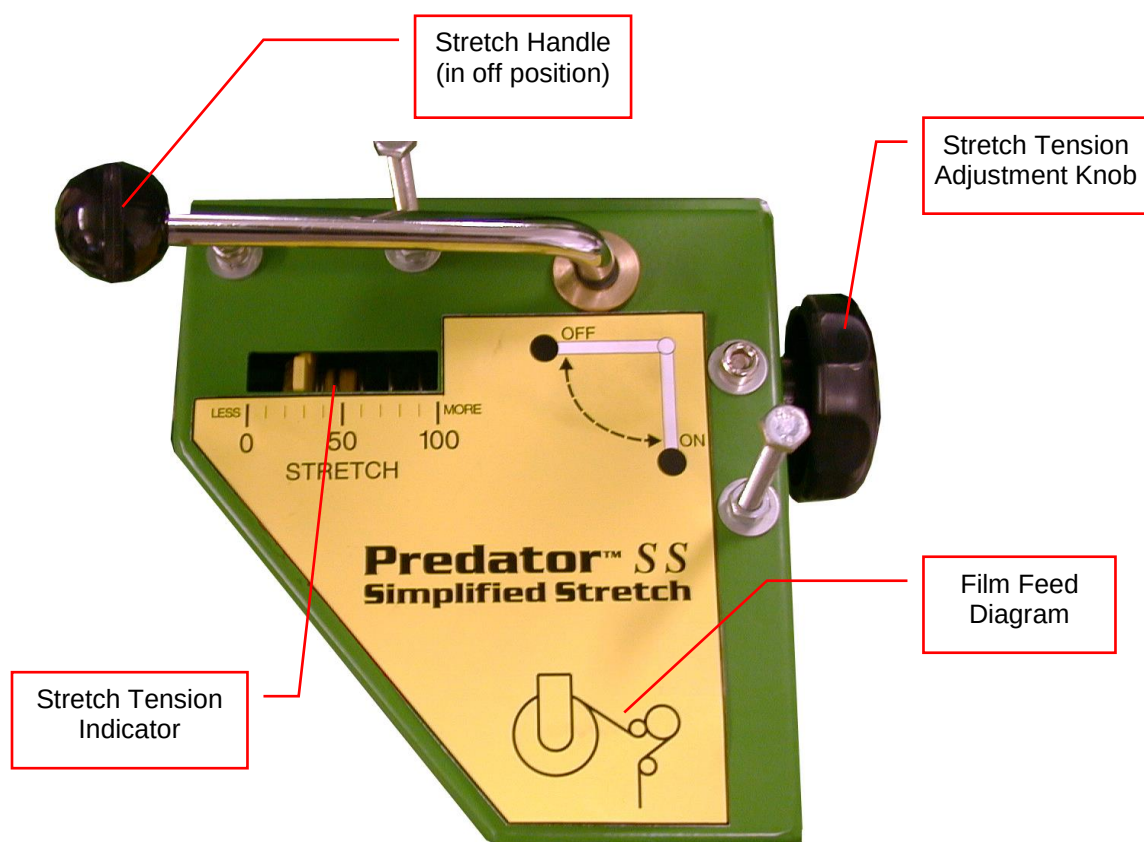


Figure 5: Film Loading/Threading Diagram

3. Follow the *Film Feed Diagram* and thread the 6-foot film tail ALL THE WAY though the rollers.
4. After threading, rotate the Stretch Tension Knob clockwise to increase film tension; or counterclockwise to decrease film tension.

5. Attach the film securely to the pallet. Tying the end of the film in a knot often helps secure the film to the pallet.

**Note**

The Stretch Tension numbers (0 to 100) DO NOT indicate the stretch percentage of your film. The numbers are used for reference only.

Machine Operating Instructions

Running a Wrap Cycle

1. To start a new wrap cycle, first place the product to be wrapped on the turntable, thread the film through the carriage, and attach the film to the product.
2. Ensure that the turntable and film carriage are at their home positions.
3. Set the desired number of Top and Bottom wraps.
4. Select the desired Wrap Mode to be applied to the product.
5. Press the Cycle Start button on the operator panel to initiate the wrap cycle.

Stop Condition

Follow the procedure below in the event of an emergency.

1. Press the “STOP” button. This cancels the current wrapping cycle and immediately stops the system.
2. Correct the problem.
3. Pull the “STOP” button out, and then perform normal system start-up procedure.



Note

After pressing the “E-STOP” button, wait for at least 60 (sixty) seconds before pulling the button back out. This will allow the microcontroller card to completely turn off.

Highlight Industries, Inc.

Predator SS Turntable Stretch Wrapper Operation Manual

Maintenance

Preventative Maintenance

As with all machinery, proper attention and maintenance is the key to long component life, maximum performance, and safe operation. By spending a few minutes reading and following these preventive measures, you should reduce the downtime and prolong the life of your system.

It is important to understand that these maintenance schedules are minimum recommendations. *Highlight Industries, Inc.* cannot possibly know, evaluate, or advise the various trades in all schedules of periodic maintenance. Accordingly, anyone who maintains or services a stretch wrap machine must first satisfy himself/herself as to the schedules of preventive maintenance based on cycling operation and environmental locations.

**Warning!**

All maintenance operations require the equipment to be powered down and locked out for personnel safety.

Lock-out/ Tag-out Procedures

Be sure that anyone performing any type of maintenance on this equipment is familiar with and is adhering to the lock-out/tag-out procedures set forth by the General OSHA or the State OSHA guidelines.

Visual Checks

Visual checks should be conducted at least once per month:

1. Keep the machine and surrounding area as clean as possible, especially near moving components.
2. Check for loose hardware, especially set screws located in: sprocket hubs, bearing hubs, and flanges.
3. Check for oil leaks around the speed reducers.
4. Check for dry seals at the bearings.
5. Check for chain wear and proper tension on the power roller stretch sprockets. The correct amount of chain tension can be checked by pulling the chain taut and having 3/8" slack.
6. Check for loose electrical connections and for frayed cords and cables. Replace immediately any damaged cords and cables.

Turntable Belt Adjustment

Refer to the figure below to adjust the turntable belt. First, loosen the four motor flange (M10) screws. Turn the (M10) bolt on the adjuster tab until the belt is tensioned. Retighten the four motor flange screws.

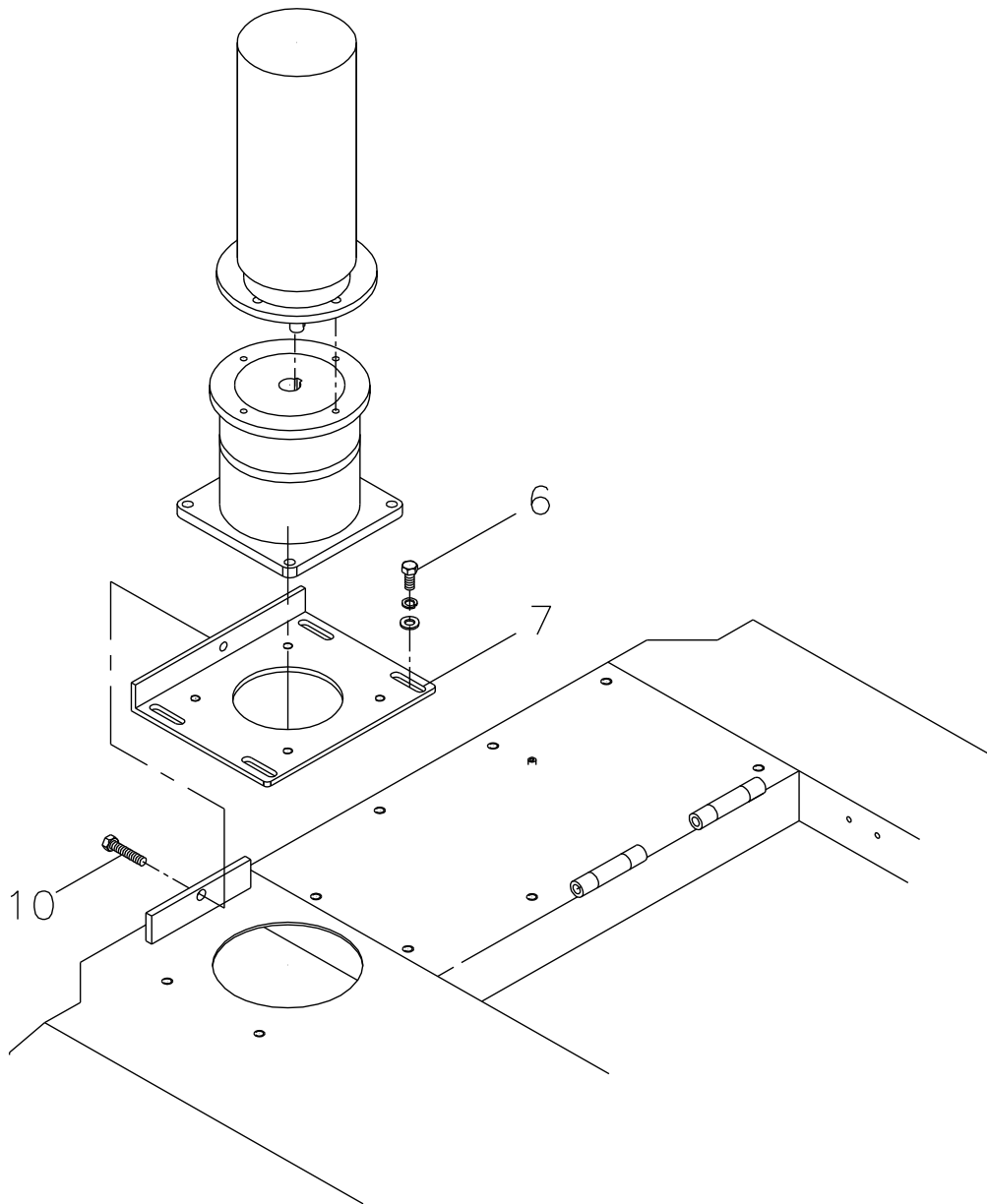


Figure 6: Turntable Belt Adjustment

Film Carriage Lift Chain Adjustment

Refer to the figure below to adjust the film carriage lift chain. First, loosen nut (M12) on the adjustment bracket. Turn the bottom bolt until desired tension is achieved. Retighten nut (M12). Note that the driving chain and adjustment bracket should be parallel with the movable pulley base.

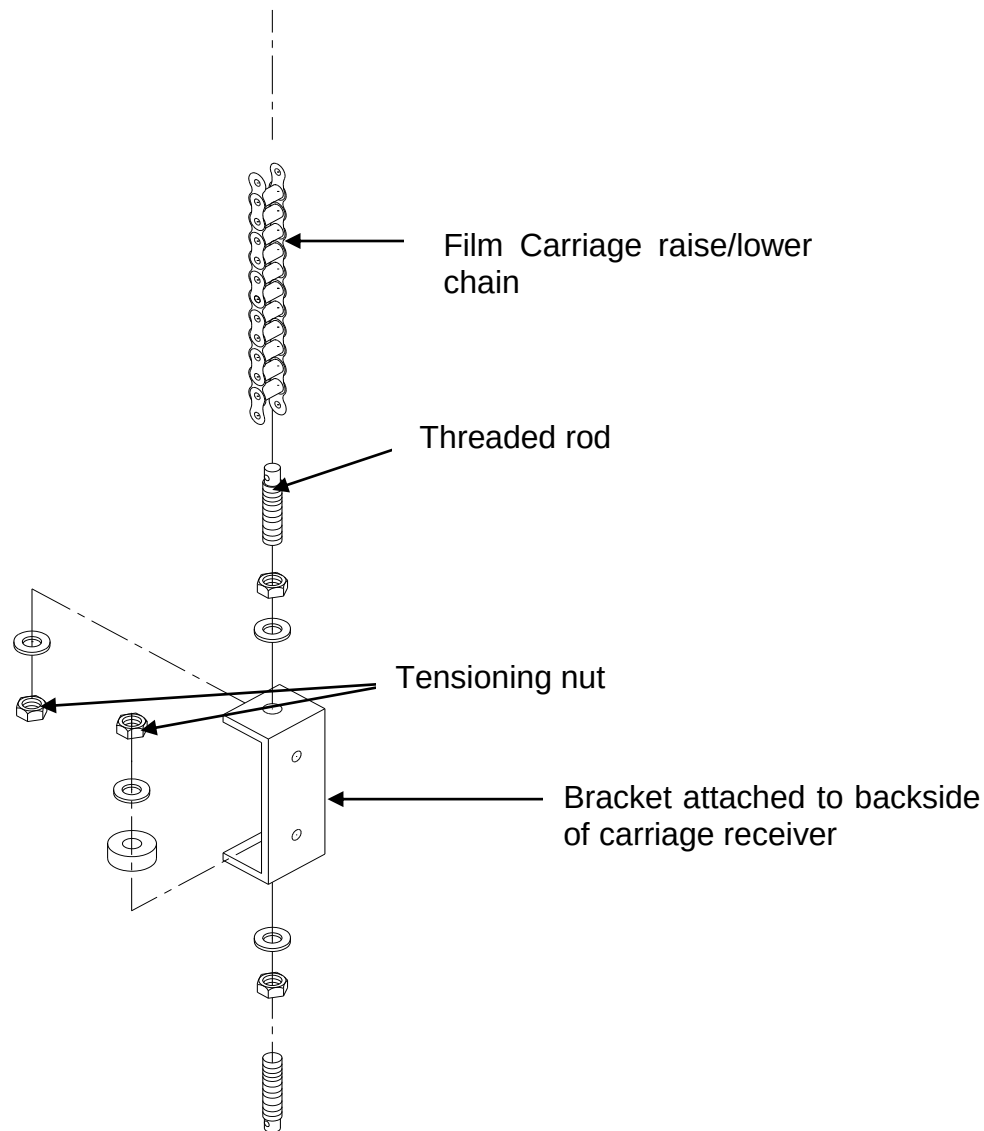


Figure 7: Film Carriage Lift Chain Adjustment

Sprockets and Chains

All sprockets should be inspected for wear, and chains should be checked for proper tension and lubrication on a periodic basis. Failure to do so will lead to premature sprocket failure. Any general-purpose chain lube should be sufficient for lubrication. Replace chain guards when preventative maintenance is complete.

Maintenance Schedule

Part	Schedule	Service With
Carriage Lift Belt	Check and tighten every six months.	
All pivot bearings	Initially first 1,000 cycles; every 10,000 cycles thereafter.	Mobil EP-2 grease

**Note**

For additional maintenance information about specific components on the machine, refer to the components operation manuals shipped inside the main enclosure.

Highlight Industries, Inc.

Predator SS Turntable Stretch Wrapper Operation Manual

Trouble- shooting

Turntable Problems

Problem	Possible Cause	Possible Solution
The turntable does not rotate manually.	The Start Button was not pressed.	The Start button needs to be pressed at the same time as the turntable job button to activate manual mode.
	The Turntable speed potentiometer is not set high enough.	Adjust the turntable speed towards 100%.
	The turntable is mechanically restricted.	Check the condition of the motor, chain, and rollers under the turntable. Replace as necessary.
	The turntable speed potentiometer has failed.	Use a multi-meter to check the resistance of the potentiometer.
	A fuse on the microcontroller card has blown.	Check fuses and replace as needed.
	The motor is not wired correctly.	Check wiring to the motor and replace as necessary.
The turntable does not stop at home.	The Turntable Home proximity switch is malfunctioning.	Check the wiring to the proximity switch and that it is adjusted properly (only 8mm sensing range). Replace as necessary.

Film Carriage Problems

Problem	Possible Cause	Possible Solution
The Film Carriage does not Raise or Lower manually or in the wrap cycle.	The Start Button was not pressed.	The Start button needs to be pressed at the same time as the carriage raise/lower button to activate manual mode.
	The Film Carriage Raised/Lowered limit switch is malfunctioning.	Ensure that the carriage limit switch is clear of any obstruction and moves smoothly. Check wiring to the limit switch and replace as necessary.
	The film carriage speed potentiometer is not set high enough.	Adjust the speed potentiometer towards 100%.
	The film carriage speed potentiometer has failed.	Use a multi-meter to check the resistance of the potentiometer.
	A fuse on the microcontroller card has blown.	Check fuses and replace as needed.
	The motor is not wired correctly.	Check wiring to the motor and replace as necessary.
The Film Carriage rises some, but does not rise to the top of the package.	The Product Height Detection photo eye is positioned too high.	Slide the Product Height Detection photo eye lower.
	The Product Height Detection photo eye does not see the load.	Black or clear loads may require an optional ultrasonic sensor instead of the standard diffuse sensor.
		Check wiring to the sensor.
		Replace the sensor as necessary.
The Film Carriage rises too far above the package.	The Product Height Detection photo eye is not adjusted properly.	Adjust position or sensitivity of the sensor.
	The Product Height Detection photo eye is positioned too low.	Slide the Product Height Detection photo eye higher.
	The Product Height Detection photo eye is not adjusted properly and always sees a 'package'.	Adjust the sensitivity of the sensor (see the Photo Sensor Adjustments and/or Pre-Operation Adjustments section).

Highlight Industries, Inc.

Predator SS Turntable Stretch Wrapper Operation Manual

Technical References

Recommended Spare Parts

The table below provides the recommended spare parts list for the Predator SS Turntable Stretch Wrapper.

Machine Section	HLI Part Number	Item Description	Qty
Turntable	300013	BELT B-172	1
Turntable	500010	PLASTIC SUPPORT WHEEL	6
Turntable	300037	TURNTABLE DRIVE MOTOR – 1/4HP (60" dia turntable)	1
Turntable	300040	TURNTABLE DRIVE MOTOR – 1/2HP (72" dia turntable)	1
Turntable	300031	TURNTABLE DRIVE REDUCER (60" dia turntable)	1
Turntable	300032	TURNTABLE DRIVE REDUCER (72" dia turntable)	1
Tower	300038	CARRIAGE LIFT MOTOR – 1/4HP	1
Tower	300030	CARRIAGE LIFT REDUCER	1

PREDATOR SS cULus RATED TURNTABLE WRAPPER

MACHINE IDENTIFICATIONS			
CUSTOMER	REPLACE	CUST	
DISTRIBUTOR	REPLACE	DIST	
PROJECT #	800-330		
SERIAL NUMBER	-		
MANUFACTURED YEAR	2023		

UTILITY SPECIFICATIONS	
1-PHASE, *EARTH	60 HERTZ
MAIN LINE VOLTAGE	120VAC, 1 PHASE
CONTROL VOLTAGE	120 VAC, 24 VDC
TOTAL MOTOR FULL LOAD	0.5 HP
FULL LOAD CURRENT	15 AMP
COMPRESSED AIR	-

WIRE REFERENCE FORMAT		
WIRE NUMBER	PAGE	LINE
0102	01	02

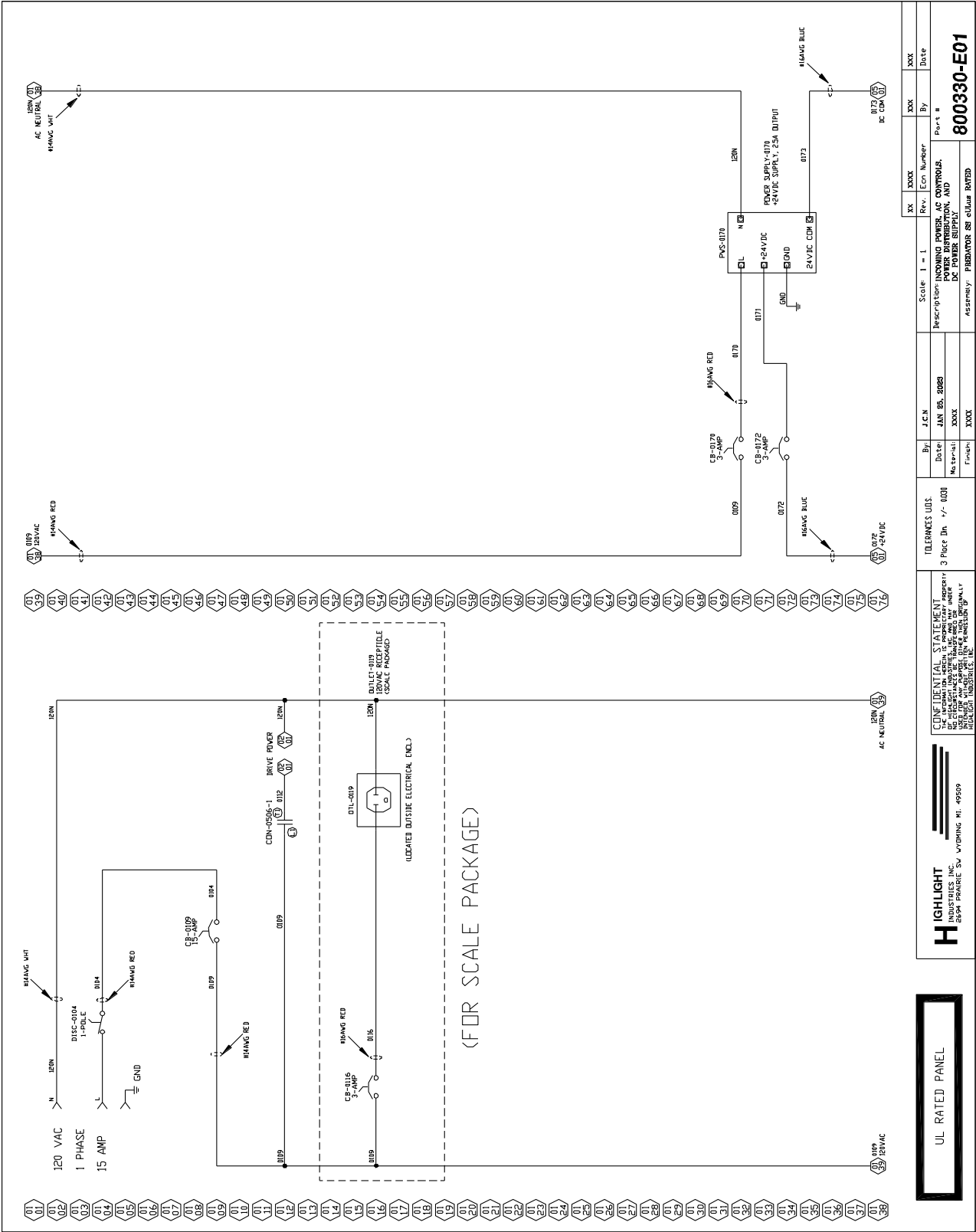
—	WIRES INTERNAL TO MAIN ENCLOSURE ONLY
---	WIRES THAT PASS OUTSIDE OF MAIN ENCLOSURE

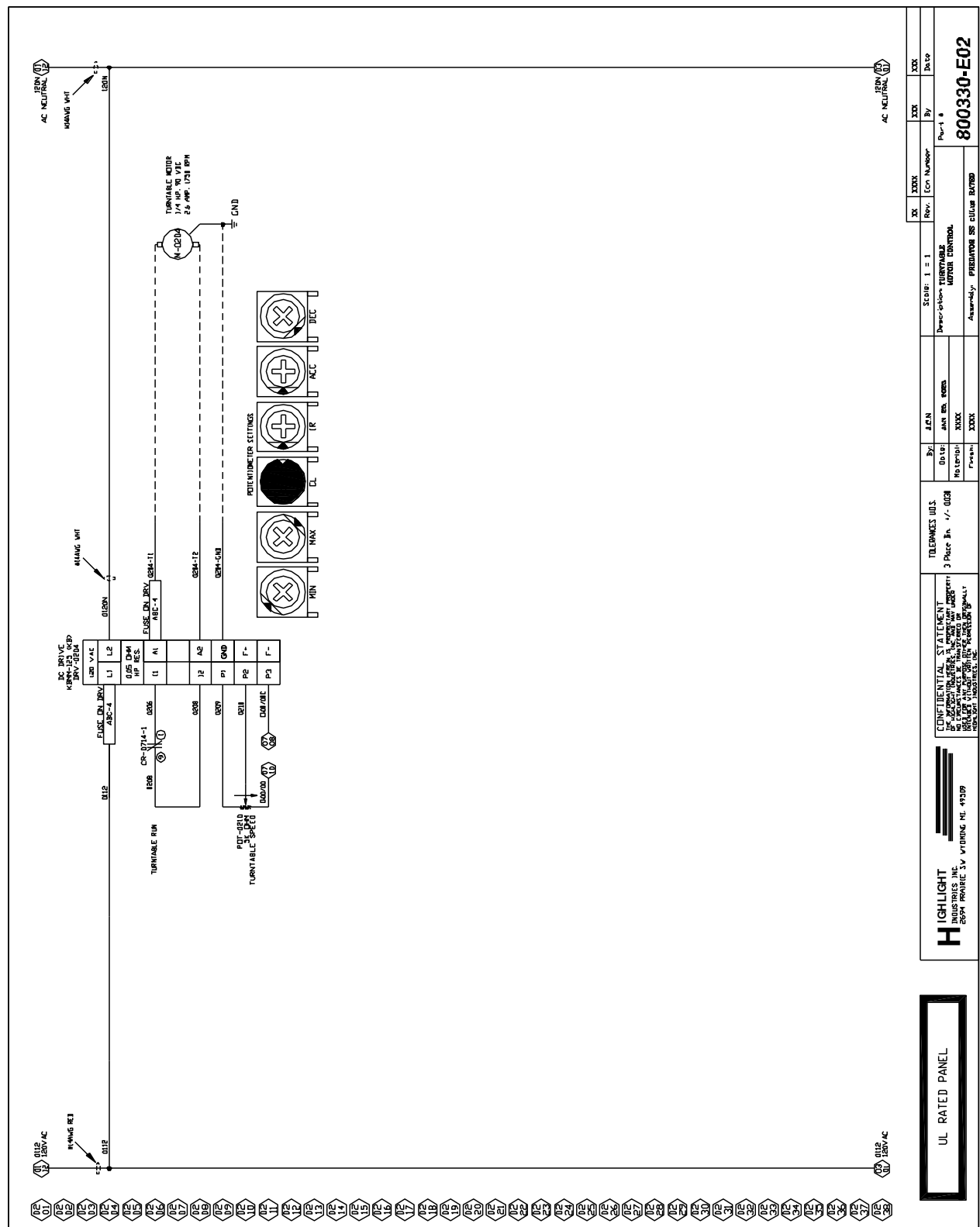
UL RATED PANEL

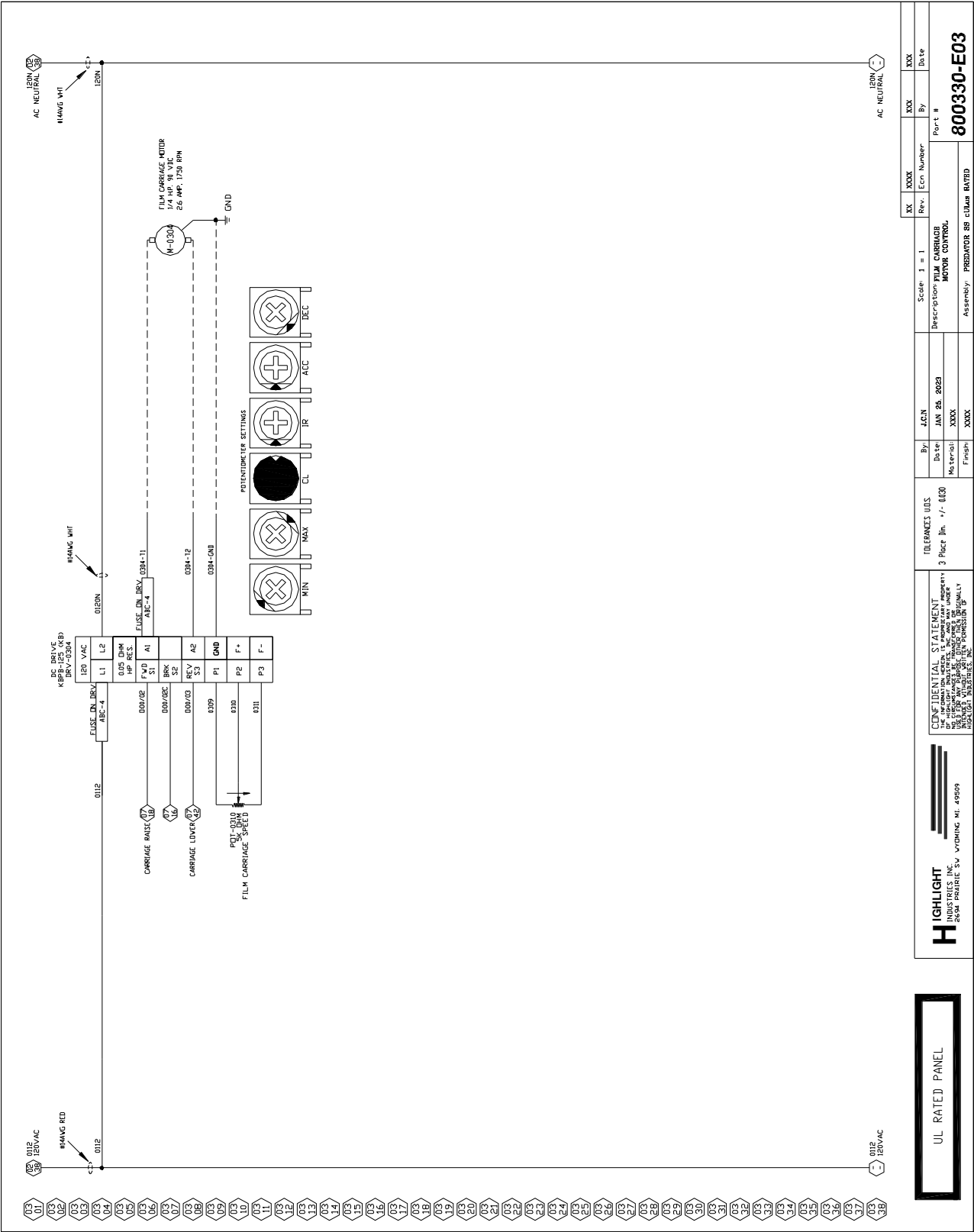
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INDUSTRIES, INC.
2694 DEARIE SW LYNNING, MI 48009

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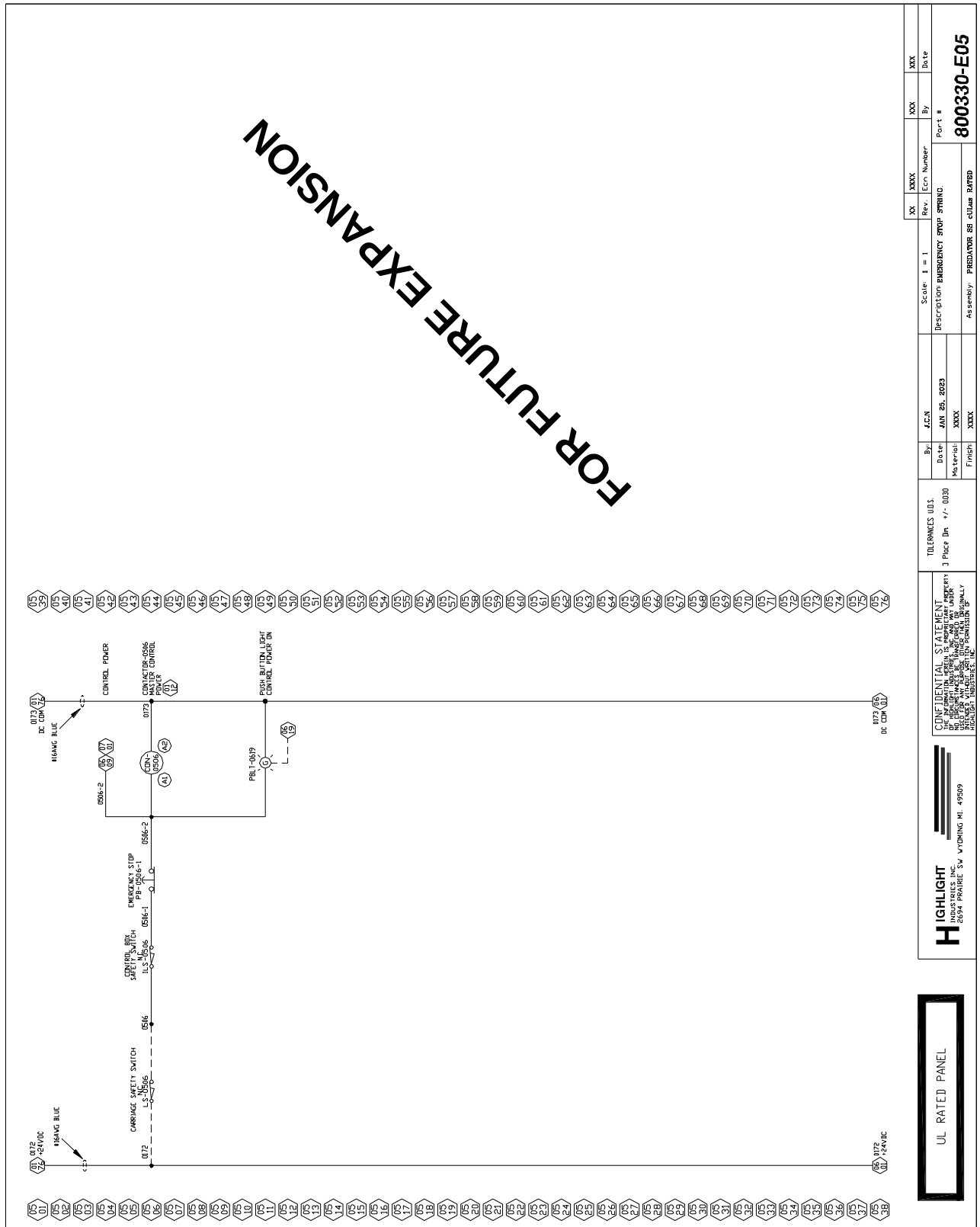
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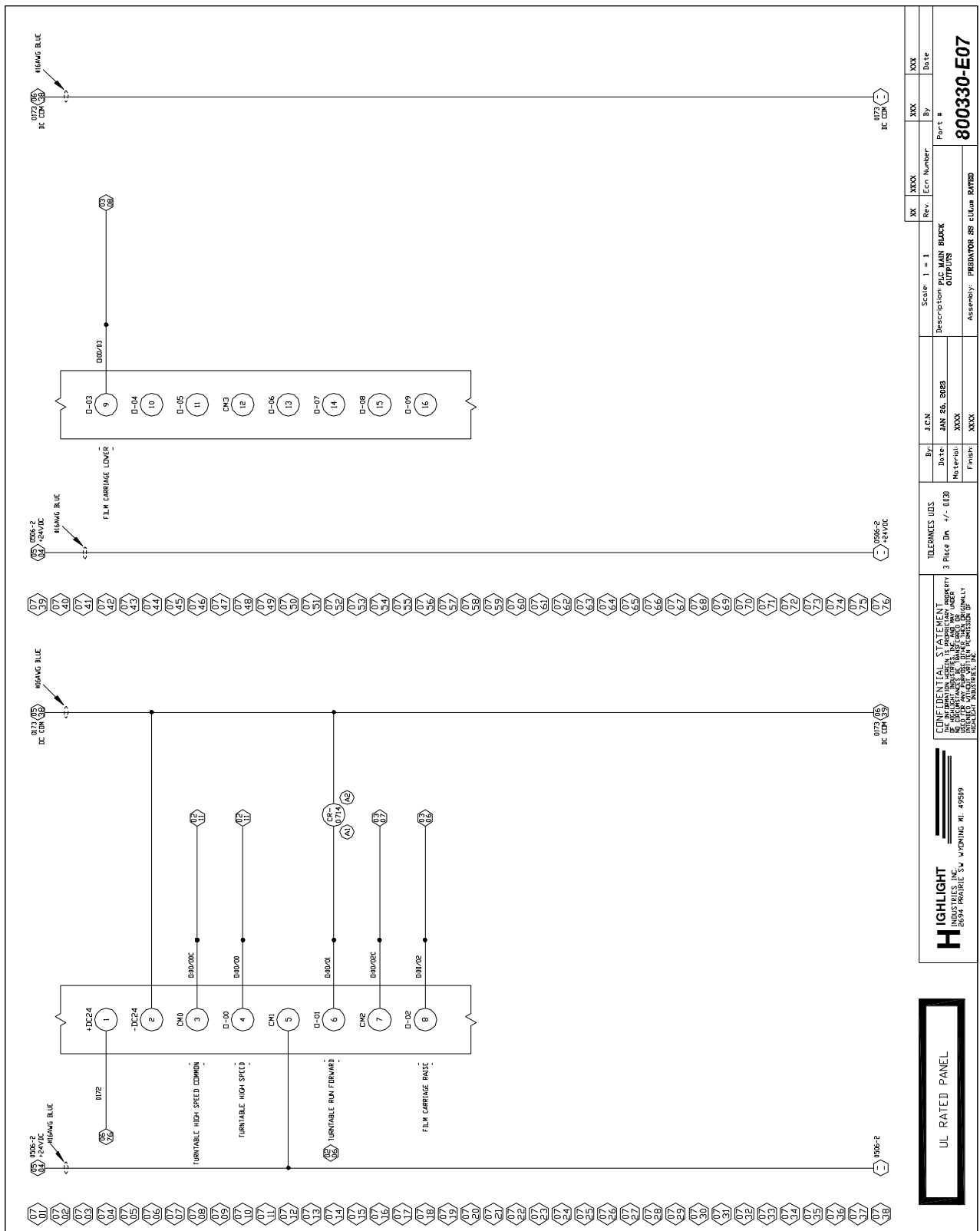


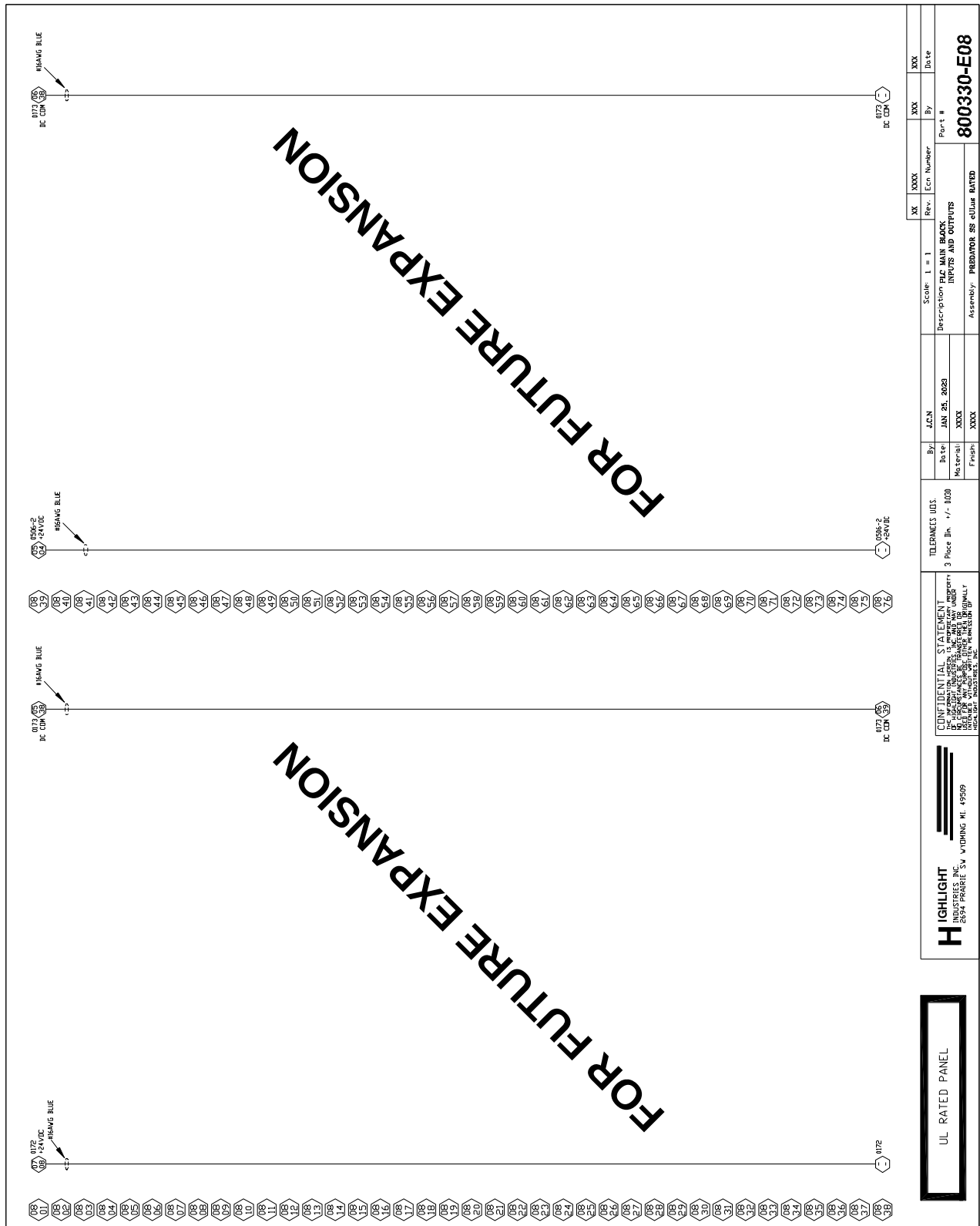




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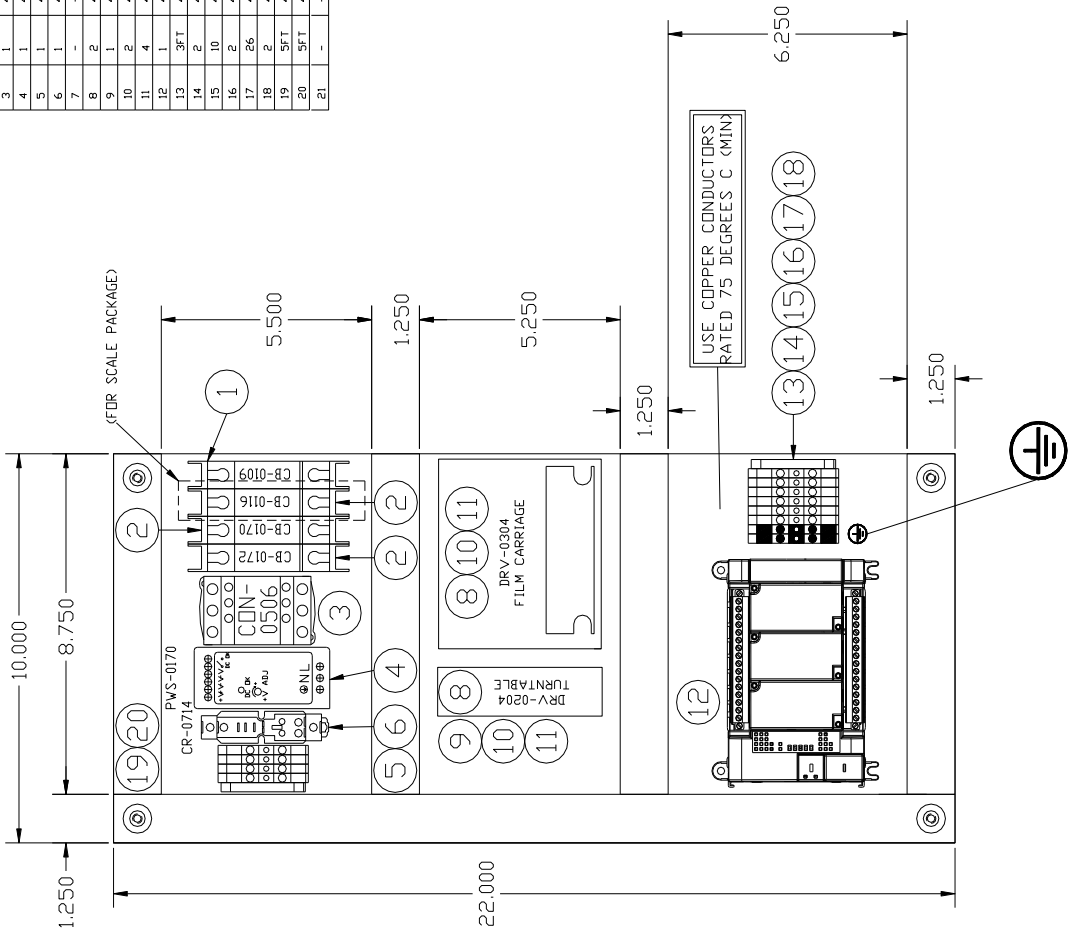
CONTROL CONNECTIONS

CABLE-0904

	CABLE #1904	
BLACK (1)	0172 -- +24VDC	
WHITE (2)	0173 -- DC COM	
RED (3)	0506 -- CARRIAGE SAFETY SWITCH	
GREEN (4)	100/02 -- FILM CARRIAGE RAISED	
ORANGE (5)	100/03 -- FILM CARRIAGE LOWERED	
YELLOW (6)	100/05 -- PKG HEIGHT DETECTION EVE	

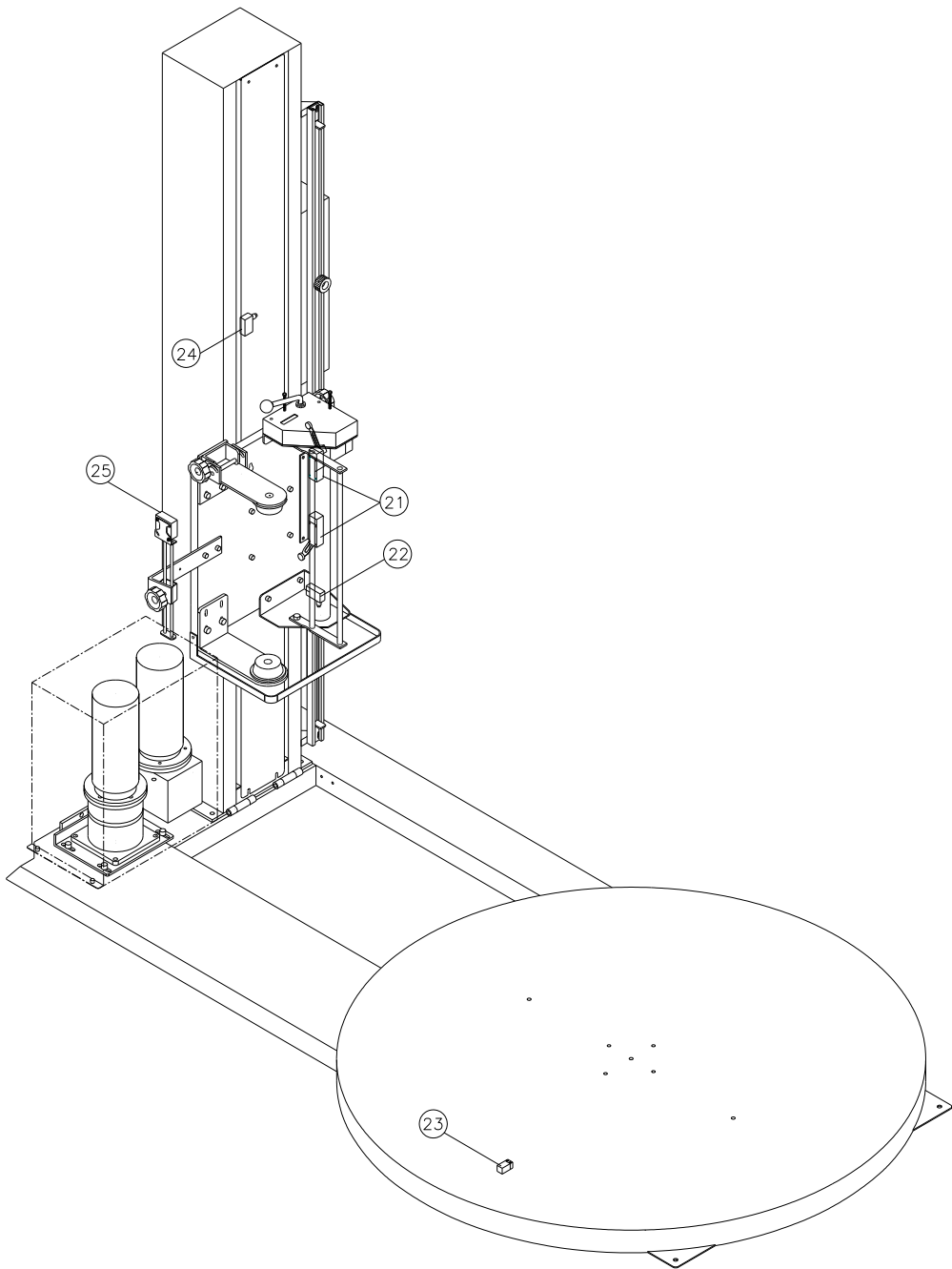
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XX	XXXX	Rev.	Ecn Number	By	Date	Assembly: PREDATOR SS c/u/aa INTD		XX	XX	XX	XX

REF.	QTY.	HLI P/N	MANUFACTURER	MFN P/N	DESCRIPTION
1	1	403982	SCHNEIDER ELECT.	60112	CIRCUIT BREAKER, 1 POLE, 15 AMP, UL 489 BRANCH, 120VAC, 60VDC
2	3	403573	SCHNEIDER ELECT.	60104	CIRCUIT BREAKER, 1 POLE, 3 AMP, UL 489 BRANCH, 120VAC, 60VDC
3	1	406100	SCHNEIDER ELECT.	LC1D18BL	CONTACTOR, 18A, 24VDC COIL, IND INC AUX
4	1	404248	MEANWELL	MDR-60-24-MW	POWER SUPPLY, 24VDC, 2.5 AMP, DIN RAIL MOUNT
5	1	406224	SCHNEIDER ELECT.	RP2F1	RELAY BASE, 1 POLE, RPM RELAYS, SPARE
6	1	406219	SCHNEIDER ELECT.	RPM2BD	RELAY, 1 POLE, FORM C, 15A, 24VDC COIL, SPARE
7	-	-	-	-	-
8	2	407385	KB ELECTRONICS	KBP8-125	DRIVE BOARD, 90 VDC, REVERSING
9	1	407241	KB ELECTRONICS	KB-9849	DRIVE BOARD FUSE KIT, LINE/ARMATURE
10	2	407237	KB ELECTRONICS	KB-9839	RESISTOR, DRIVE BOARD HP RESISTOR, 0.05 OHM, 1/4HP@90V
11	4	405824	BUSMAN	ABC-4	FUSE, 4 AMP, ABC
12	1	406849	ALLEN-BRADLEY	2098-LC0-240V8	MICRO 830 CONTROLLER
13	3FT	407249	Drilon	PFP-100N	DIN RAIL MOUNT CHANNEL
14	2	408227	ALLEN-BRADLEY	1492-ERC35	TERMINAL BLOCK, END ANCHOR
15	10	408243	ALLEN-BRADLEY	1492-J4	TERMINAL BLOCK, 600V, 35A
16	2	408328	ALLEN-BRADLEY	1492-CBJ3	TERMINAL BLOCK, END PLATE
17	26	408544	ALLEN-BRADLEY	1492-M6x12	MARKING TAG FOR J-4 SERIES TERMINAL BLOCK
18	2	408545	ALLEN-BRADLEY	1492-JG4	TERMINAL BLOCK, GROUND, 600V, 35A
19	5FT	405503	PANDUIT	G1X2L06	WIRE DUCT, 1" X 2"
20	5FT	405504	PANDUIT	C1L06	WIRE DUCT COVER, 1"
21	-	-	-	-	-



UL RATED PANEL	HIGHLIGHT INDUSTRIES INC. 2694 PRAIRIE SW WYOMING MI 49509	CONFIDENTIAL STATEMENT OF DESIGN, INVENTION, AND MANUFACTURE THIS DOCUMENT CONTAINS PROPRIETARY INFORMATION WHICH IS UNLAWFULLY DISCLOSED TO THE PUBLIC BY HIGHLIGHT INDUSTRIES, INC.	TOLERANCES UNLESS SPECIFIED		By: J.C.N. Date: JAN 26, 2023		Scale: 1 = 1		Rev: XXXX Ecn Number: XXXX		XX XXXX XXX XXX	
			3 Place Dns +/- .0030		Description: ELECTRICAL SUBPANEL LAYOUT		Part #		By: XXXX		Date: XXXX	
							800331-SP		Assembly: PREDATOR SS cULus RATED			

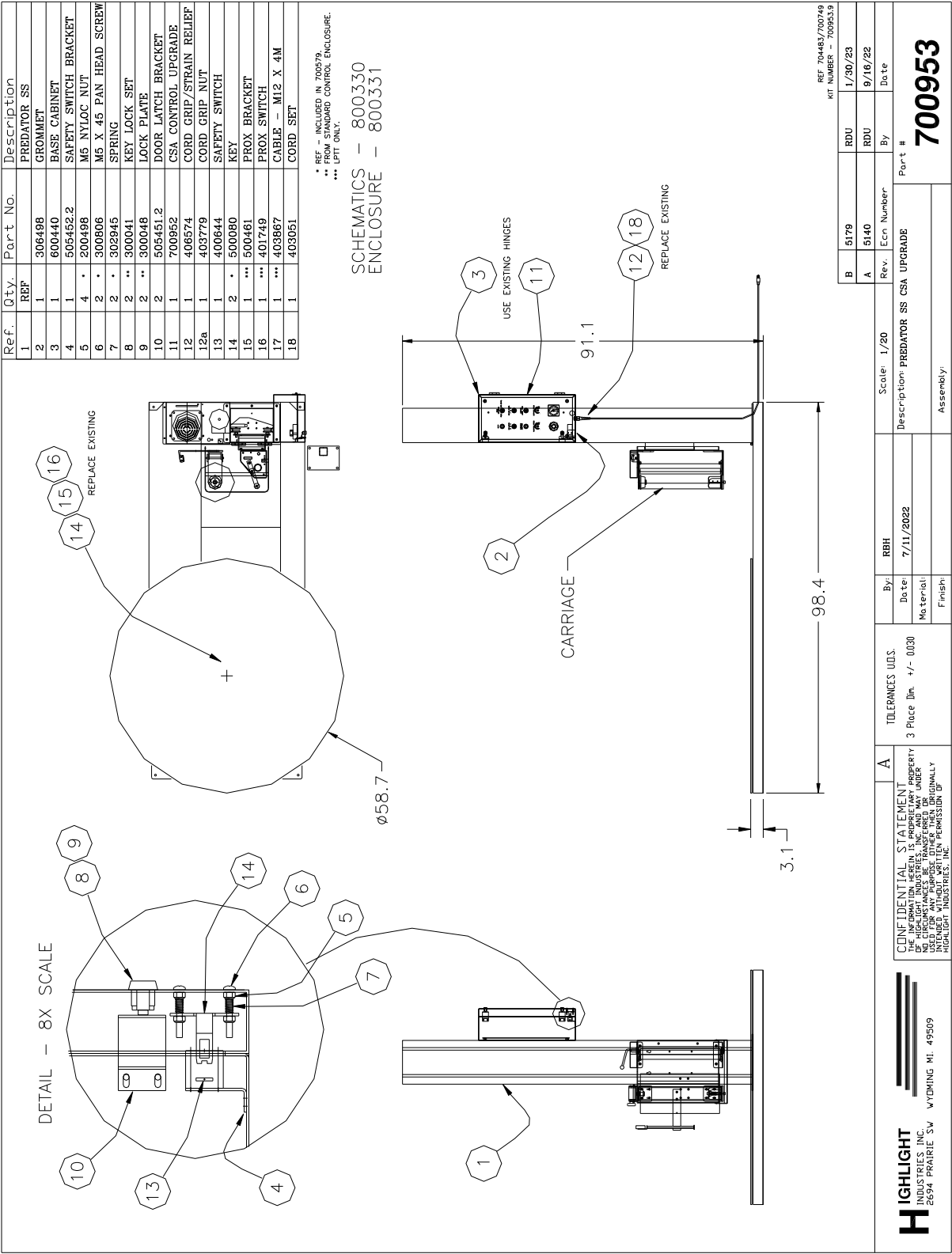
Sensor Layout



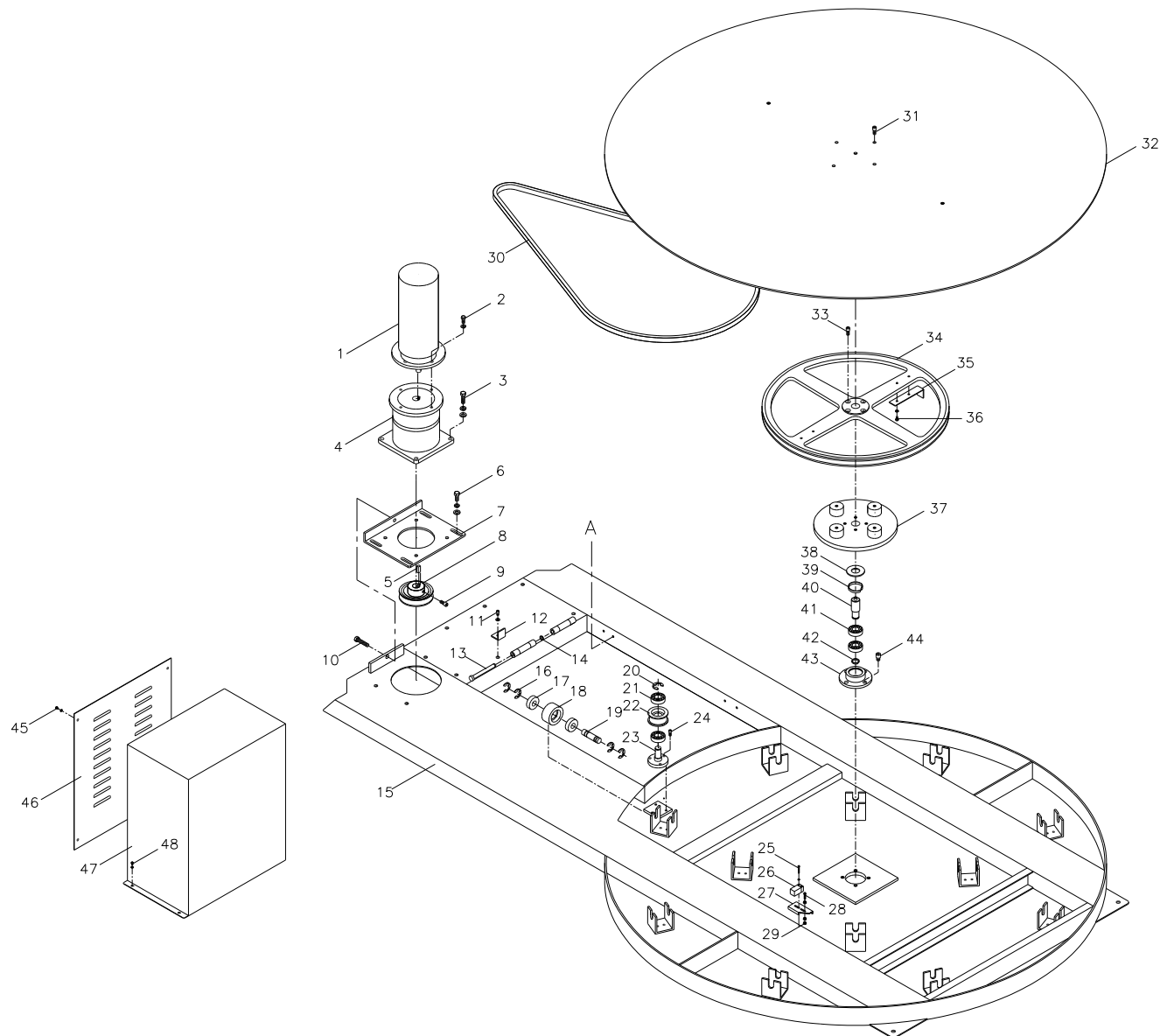
Machine Section	HLI Part Number	Item Description	Qty
21	400639	LIMIT SWITCH TZ-8108	2
22	400636	LIMIT SWITCH TM-1307	1

23	401749	PROXIMITY SWITCH	1
24	400637	LIMIT SWITCH TZ-7110	1
25	400643	PHOTO SENSOR A3R-2MX(OPTION)	1

Mechanical Drawings



Turtable Frame Assembly

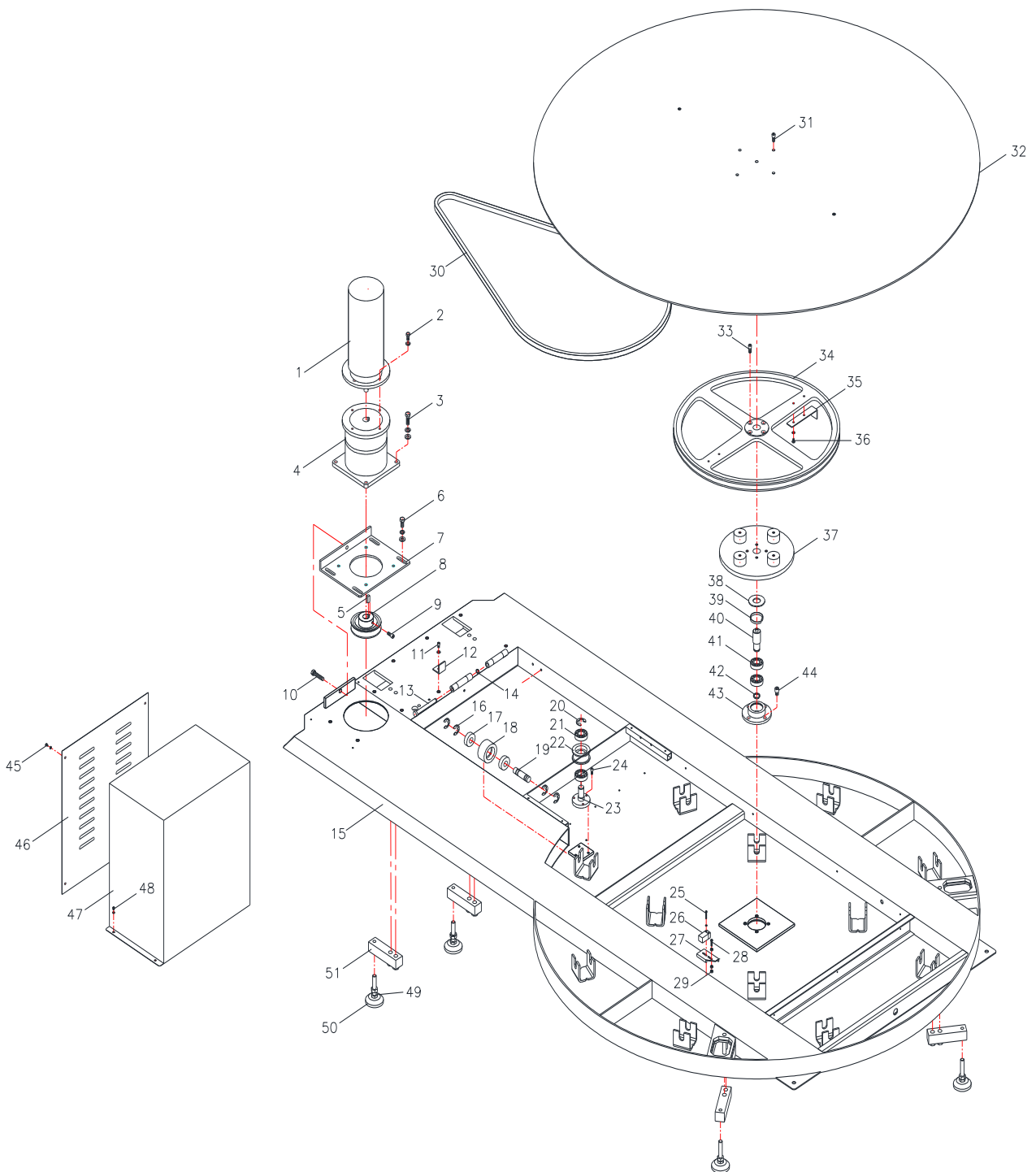


Turndtable Parts List

Machine Section	HLI Part Number	Item Description	Qty
1	300037	MOTOR VP3310D 1/4HP →1.5M	1
	300040	MOTOR VP3330D 1/2HP →1.8M	1
2	200091	SCREW M8X20L & W.	
3	200092	SCREW M10x20L & S.W. & W.	4
4	300031	SPEED REDUCER 1/4HP →1.5M	1
	300032	SPEED REDUCER 1/2HP →1.8M	1
5	300002	KEY 5X20L	1
6	200093	SCREW M10x25L & S.W. & W.	4
7	500034	MOTOR ADJUST SEAT →1.5M	1
	500035	MOTOR ADJUST SEAT →1.8M	1
8	500014	MOTOR PULLEY 4"22 →1.5M	1
	500015	MOTOR PULLEY 4"28 →1.8M	1
9	207303	SET SCREW, M8 x 10L	2
10	200095	SCREW M10x70L	1
11	200077	SCREW M6x12L & W.	1
12	500083	MOTOR BOX FIX PLATE	1
13	500087	PIN	2
14	300006	RING STW-10	2
15	600000	TURNTABLE SEAT 1.5M	1
	600001	TURNTABLE SEAT 1.8M	1
16	300009	RING STW-17	48
17	300021	BEARING 6203ZZ →1.5M	24
	300021	BEARING 6203ZZ →1.8M	28
18	500010	SUPPORTING PLASTIC WHEEL →1.5M	12
	500010	SUPPORTING PLASTIC WHEEL →1.8M	14
19	500009	SUPPORTING PLASTIC WHEEL SHAFT →1.5M	12
	500009	SUPPORTING PLASTIC WHEEL SHAFT →1.8M	14
20	300009	RING STW-17	1
21	300021	BEARING 6203ZZ	2
22	500011	BELT LOCK PULLEY	1
23	500012	BELT LOCK PULLEY SHAFT	1
24	200078	SCREW M6x16L	3
25	200098	SCREW M3x25L & W.	2
26	400641	PROXIMITY SWITCH PS-05N-3M	1

27	500013	PROXIMITY SWITCH FIX SEAT	1
28	200103	SCREW M4x20L	2
29	200128	NUT M5	6
30	300013	BELT B-172 →1.5M	1
	300014	BELT B-183 →1.8M	1
31	200113	SCREW M8x25L	4
32	500004	TURNTABLE PLATE 1.5M	1
	500005	TURNTABLE PLATE 1.8M	1
33	200084	SCREW M8x20L	4
34	500007	WHEEL 22"	1
35	500006	LOCATING PLATE	1
36	200108	SCREW M5x10L & W.	4
37	500008	PULLEY FIXING PLATE	1
38	500085	BELT PULLEY SEAT RING	1
39	300012	RING RTW-42	1
40	500084	BELT PULLEY SHAFT	1
41	300018	BEARING 6004ZZ	2
42	300010	RING STW-20	1
43	500086	BELT PULLEY BEARING SEAT	1
44	200088	SCREW M10x16L	4
45	200100	SCREW M4x8L & W.	4
46	500081	MOTOR COVER →1.5M	1
	500082	MOTOR COVER →1.8M	1
47	600028	TURNTABLE MOTOR BOX 1.5M	1
	600029	TURNTABLE MOTOR BOX 1.8M	1
48	200100	SCREW M4x8L & W.	2
49	600038	RAMP (FOR 60" TURNTABLE - NOT SHOWN)	OPT
50	600039	RAMP PORCH – 24" (FOR 60" TURNTABLE - NOT SHOWN)	OPT
51	602297	RAMP PORCH – 48" (FOR 60" TURNTABLE - NOT SHOWN)	OPT

Turntable Scale Frame Assembly (optional)

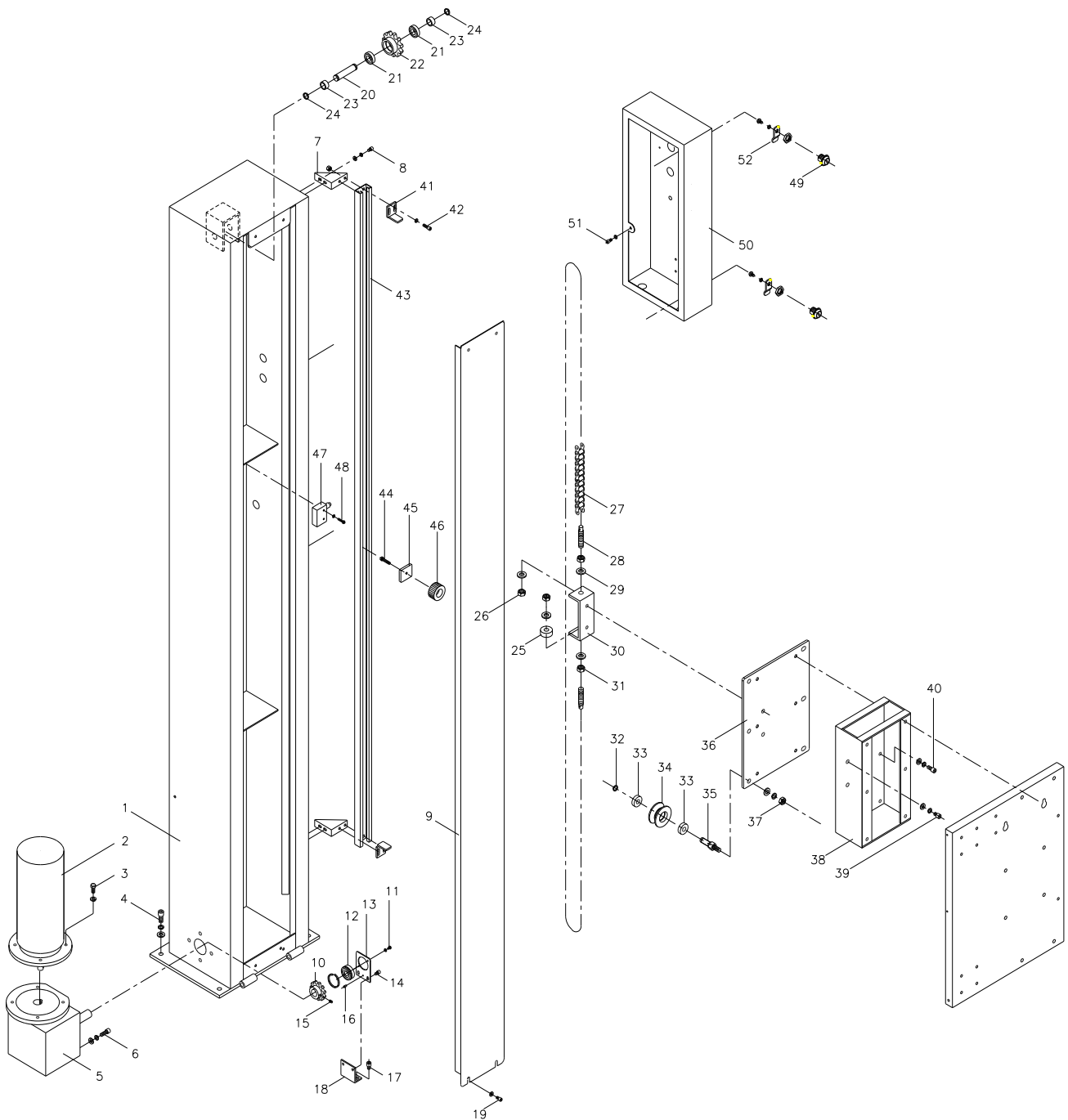


Turntable Parts List

Machine Section	HLI Part Number	Item Description	Qty
1	300037	MOTOR VP3310D 1/4HP →1.5M	1
2	200091	SCREW M8X20L & W.	
3	200092	SCREW M10x20L & S.W. & W.	4
4	300031	SPEED REDUCER 1/4HP →1.5M	1
5	300002	KEY 5X20L	1
6	200093	SCREW M10x25L & S.W. & W.	4
7	500034	MOTOR ADJUST SEAT →1.5M	1
8	500014	MOTOR PULLEY 4"22 →1.5M	1
9	207303	SET SCREW, M8 x 10L	2
10	200095	SCREW M10x70L	1
11	200077	SCREW M6x12L & W.	1
12	500083	MOTOR BOX FIX PLATE	1
13	500087	PIN	2
14	300006	RING STW-10	2
15	600424	TURNTABLE SEAT WITH INTEGRAL SCALE - 1.5M	1
16	300009	RING STW-17	48
17	300021	BEARING 6203ZZ →1.5M	24
18	500010	SUPPORTING PLASTIC WHEEL →1.5M	12
19	500009	SUPPORTING PLASTIC WHEEL SHAFT →1.5M	12
20	300009	RING STW-17	1
21	300021	BEARING 6203ZZ	2
22	500011	BELT LOCK PULLEY	1
23	500012	BELT LOCK PULLEY SHAFT	1
24	200078	SCREW M6x16L	3
25	200098	SCREW M3x25L & W.	2
26	400641	PROXIMITY SWITCH PS-05N-3M	1
27	504280	PROXIMITY SWITCH FIX SEAT	1
28	200103	SCREW M4x20L	2
29	200128	NUT M5	6
30	300013	BELT B-172 →1.5M	1
31	200113	SCREW M8x25L	4
32	500004	TURNTABLE TOP →1.5M	1

33	200084	SCREW M8x20L	4
34	500007	WHEEL 22"	1
35	504281	LOCATING PLATE	1
36	200108	SCREW M5x10L & W.	4
37	500008	PULLEY FIXING PLATE	1
38	500085	BELT PULLEY SEAT RING	1
39	300012	RING RTW-42	1
40	500084	BELT PULLEY SHAFT	1
41	300018	BEARING 6004ZZ	2
42	300010	RING STW-20	1
43	500086	BELT PULLEY BEARING SEAT	1
44	200088	SCREW M10x16L	4
45	200100	SCREW M4x8L & W.	4
46	500081	MOTOR COVER →1.5M	1
47	600028	TURNTABLE MOTOR BOX 1.5M	1
48	200100	SCREW M4x8L & W.	2
49	200140	JAM NUT – M12	REF
50	306276	LOAD CELL FOOT	4
51	306275	LOAD CELL	4
52	300353	SCALE DISPLAY (NOT SHOWN)	1
53	600947	RAMP – INTEGRATED SCALE (NOT SHOWN)	OPT

Tower Assembly

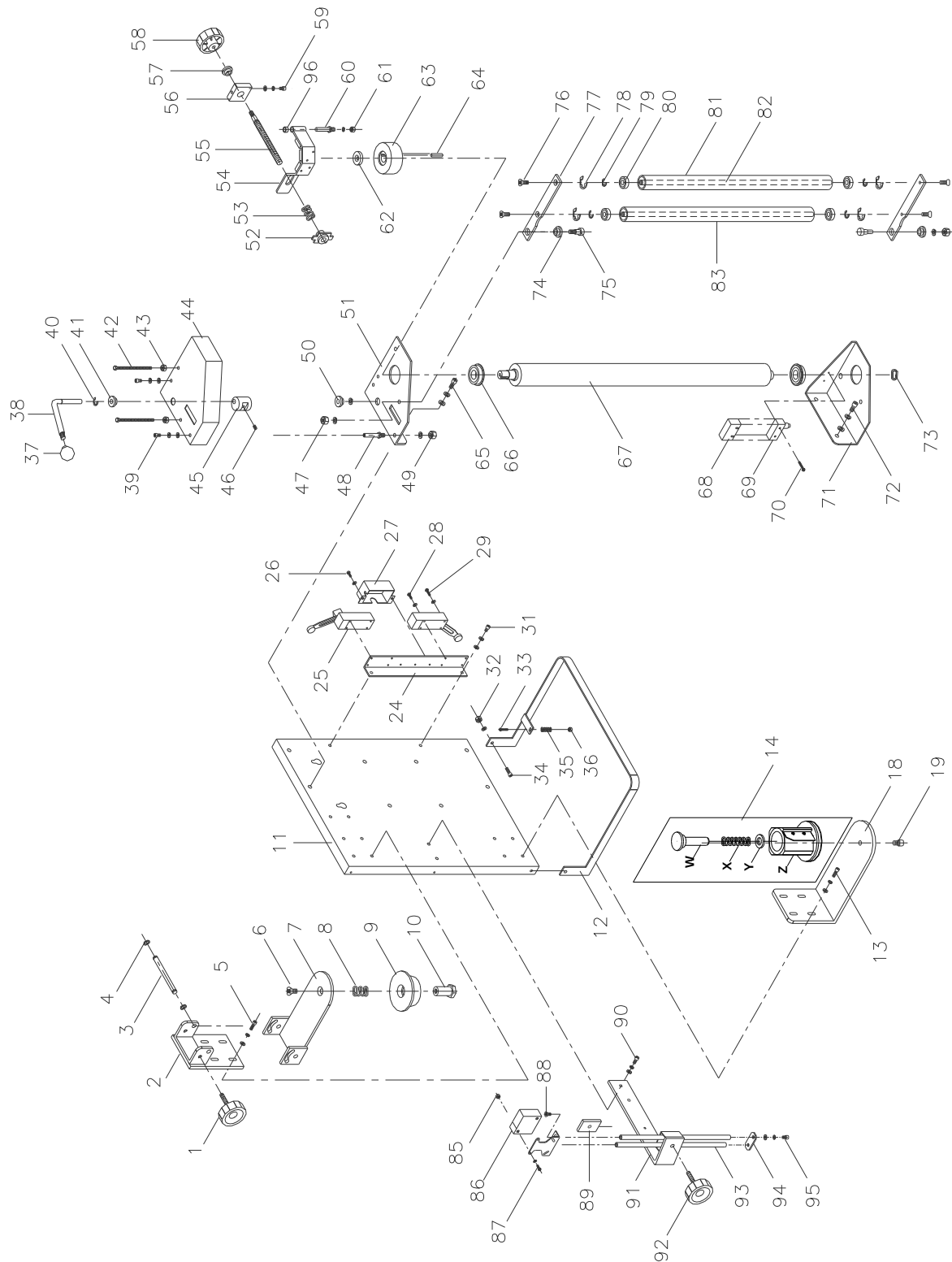


Tower Parts List

Machine Section	HLI Part Number	Item Description	Qty
1	500133	TOWER - STD	1
	500137	TOWER - 110	1
2	300038	ELEVATOR UNIT MOTOR VP3310D	1
3	200091	SCREW M8x20L & S.W.	4
4	200090	SCREW M10x25L & S.W. & W.	5
5	300030	ELEVATOR UNIT SPEED REDUCER VTM-40	1
6	200085	SCREW M8x25L & S.W. & W.	4
7	500135	CHANNEL SUPPORT	2
8	200078	SCREW M6x16L & S.W. & W.	4
9	500136	CHAIN WHEEL COVER - STD	1
	500290	CHAIN WHEEL COVER - 110	1
10	500090	UP & DOWN CHAIN WHEEL (LOWER) 11 TEETH	1
11	200099	SCREW M4x6L & W.	2
12	300020	BEARING 6202-2NR	1
13	500102	SPEED REDUCER BEARING SEAT	1
14	200076	SCREW M6x10L & W.	2
15	200120	SCREW M6x8L	1
16	300000	PIN Ø4X8	1
17	200082	SCREW M8x12L	2
18	500103	BEARING ADJUST SEAT	1
19	200077	SCREW M6x12L & W.	4
20	500003	UPPER CHAIN WHEEL SHAFT	1
21	300017	BEARING 6002ZZ	2
22	500089	UPPER CHAIN WHEEL	1
23	500093	RING	2
24	300008	RING STW-15	2
25	500104	ADJUST BAR WASHER	1
26	200140	NUT M12	1
27	300049	CHAIN - STD	1
	500488	CHAIN - 110	1
28	500025	ELEVATOR UNIT CHAIN SCREW SHAFT	2
29	200147	WASHER M12	4
30	500092	CHAIN ADJUST SEAT	1

31	200133	NUT M12	3
32	300007	RING STW-12	6
33	300016	BEARING 6001ZZ	8
34	500090	UP & DOWN DRIVING WHEEL	4
35	500091	UP & DOWN DRIVING WHEEL SHAFT	4
36	500040	CHAIN WHEEL SEAT	1
37	200131	NUT M10 & S.W. & W.	6
38	600008	ELEVATOR UNIT SEAT (INNER)	1
39	200084	SCREW M8x20L & S.W. & W.	2
40	200083	SCREW M8x16L & S.W. & W.	6
41	500138	LIMIT PLATE	2
42	200125	SCREW M6x16L & S.W & NUT	4
43	500139	CHANNEL - STD	1
	500140	CHANNEL - 110	1
44	200078	SCREW M6x16L	1
45	500141	FIX PIECE	1
46	500142	HANDEL WHEEL	1
47	400637	LIMIT SWITCH TZ-7110	1
48	200105	SCREW M4x30L & W	2
49	300041	LOCK w/ KEY	2
50	500143	CONTROL BOX	1
50a	310008	BUTTON MEMBRANE LABEL FOR CONTROL PANEL	1
51	200076	SCREW M6x10L & S.W.	1
52	300048	LOCK LATCH PLATE	2

Film Carriage Assembly



Film Carriage Parts List

Machine Section	HLI Part Number	Item Description	Qty
1	500000	HANDLE	1
2	500144	FILM FIX SEAT(UPPER)	1
3	500145	SHAFT	1
4	300006	RING STW-10	2
5	200079	SCREW M6*20L & S.W. & W.	4
6	200089	SCREW M10*20L	1
7	500146	FILM MOVING SHAFT	1
8	500147	FILM SEAT SPRING	1
9	500148	FILM SEAT(UPPER)	1
10	500101	FILM SEAT SHAFT	1
11	500149	ELEVATOR UNIT SEAT(OUTER)-20"	1
	504260	ELEVATOR UNIT SEAT(OUTER)-30"	1
12	500150	PEDAL	1
13	200079	SCREW M6*20L & S.W. & W.	4
14	700072	LOWER CORE HOLDER ASSEMBLY	1
	W	FILM SEAT SHAFT	
	X	SPRING	
	Y	BEARING NTB/A51730	
	Z – 500107	FILM FIX SEAT	
	Za – 500111	BRAKE	
	Zb – 500109	FILM BRAKE PLATE	
	Zd – 500112	WASHER	
18	500151	FILM FIX SEAT(LOWER)	1
19	200089	SCREW M10*20L	1
23	200079	SCREW M6*20L & S.W. & W.	2
24	500152	LIMIT SWITCH SEAT	1
25	400639	LIMIT SWITCH TZ-8108	2
26	200101	SCREW M4*12L	2
27	500130	TERMINAL COVER	1
28	200102	SCREW M4*16L	4
29	200101	SCREW M4*12L	4
31	200077	SCREW M6*12L & S.W. & W.	2
32	200137	NUT M6	2

33	200103	SCREW M4*20L	1
34	200079	SCREW M6*20L	2
35	500016	SPRING	1
36	200127	NUT M4	2
37	500913	HANDLE KNOB	1
38	500153	HANDLE	1
39	200077	SCREW M6*12L & S.W. & W.	2
40	300006	RING STW-10	1
41	500154	RING	1
42	200157	SCREW M6*100L	2
43	200129	NUT M6	4
44	500155	STRETCH SEAT COVER(UPPE)	1
45	500156	WHEEL	1
46	200121	SCREW M6*10L	1
47	200130	NUT M8 & S.W.	2
48	500157	STRETCH SEAT SHAFT	1
49	200130	NUT M8 & W.	1
50	500158	RING	1
51	500159	UPPER STRETCH SEAT	1
52	500160	TENSION ADJUST SEAT	1
53	500161	STRETCH SPRING	1
54	500162	BRAKE PLATE	1
55	500163	TENSION ADJUST SCREW	1
56	500164	GEAR SCREW FIX SEAT	1
57	500154	RING	1
58	500000	HANDLE	1
59	200077	SCREW M6*12L & S.W. & W.	2
60	500165	BREAK PLATE SHAFT	1
61	200129	NUT M6 & S.W.	2
62	500166	BREAK WHEEL FIX SHAFT	1
63	500167	BRAKE WHEEL	1
64	300002	PIN 5*5*20L	1
65	200084	SCREW M8*20L & S.W. & W.	2
66	300051	BEARING 6004ZZ-NR	2
67	500168	PRE-STRETCH SHAFT-20"	1
	504261	PRE-STRETCH SHAFT-30"	1

68	400640	WIRE PROTECTOR	1
69	400636	LIMIT SWITCH TM-1307	1
70	200104	SCREW M4*25L	2
71	500169	LOWER STRETCH SEAT	1
72	200084	SCREW M8*20L & S.W. & W.	2
73	300010	RING STW-20	1
74	500158	RING	2
75	500170	MOVING SHAFT PIN	2
76	200833	SCREW M6*12L	4
77	500171	MOVING SHAFT	1
	500172	MOVING SHAFT	1
78	300052	RING STW-22	4
79	300006	RING STW-10	4
80	300015	BEARING 6000ZZ	4
81	500173	IDLER ROLLER-20"	1
	504262	IDLER ROLLER-30"	1
82	500174	IDLER ROLLER SHAFT-20"	2
	504263	IDLER ROLLER SHAFT-30"	2
83	500175	IDLER ROLLER-20"	1
	504264	IDLER ROLLER-30"	1
85	200127	NUT M4 (OPTION)	2
86	400643	PHOTO SENSOR A3R-2MX (OPTION)	1
87	200101	SCREW M4x12L & W. (OPTION)	2
88	200110	SCREW M6x12L (OPTION)	2
89	500094	FIX PIECE (OPTION)	1
90	200079	SCREW M6*20L & S.W. & W. (OPTION)	2
91	500176	PHOTO SENSOR ADJUST SEAT (OPTION)	1
92	500000	HANDLER (OPTION)	1
93	500095	PHOTO SENSOR ADJUST BAR (OPTION)	2
94	500096	CONNECTING PLATE (OPTION)	1
95	200077	SCREW M6*12L & S.W. & W. (OPTION)	2
96	304629	SPACER, ANTI-SQUEAK	1

Advanced Infrared Eye - Optional

Ref.	Qty.	Part No.	Description
1	1	540879	EYE BRACKET
2	1	409305	ADVANCED EYE-300CM
3	1	405011	CABLE - MB X 4M
4	3	407218	BUTT CONNECTOR

1) REMOVE EXISTING EYE BRACKET FROM THE TOP OF THE TWO VERTICAL RODS. THE EXISTING EYE MAY TO BE REMOVED FROM BRACKET TO REVEAL SCREW HEADS.

2) MOUNT UPGRADE EYE BRACKET ON TOP OF 2 RODS USING THE EXISTING HARDWARE.

3) ROUTE NEW CABLE TO THE BOX WHERE THE EXISTING EYE WIRING IS TERMINATED FOLLOWING THE ROUTE OF THE EXISTING CABLE. REMOVE THE COVER.

4) CUT OFF EXISTING, GREY EYE CABLE AND BARE WIRES FROM THE MACHINE SIDE.

5) CONNECT THE NEWLY BARED WIRES FROM THE EXISTING GREY CABLE TO THE WIRES IN THE UPGRADE EYE CABLE USING THE WIRING SCHEDULE BELOW. CRIMP WITH SUPPLIED FITTINGS.

6) REPLACE THE WIRING COVER.

A		RDU		Scale: 1/8		Rev.		By		Date	
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HIGHLIGHT INDUSTRIES, INC. 2694 PRAIRIE SW WYOMING MI. 49509		Date: 12/27/22		Material: LAMBAO		Assembly		740088			

Highlight Industries, Inc.

Predator SS Turntable Stretch Wrapper Operation Manual

Notes

This image shows a single sheet of white paper with horizontal blue ruling lines. The lines are evenly spaced and run across the width of the page. There are no margins, text, or other markings on the paper.

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